Can Power Draw Constrain Covert Compute?

Tom Kimpson [1] **Mauricio Baker** [2] **Emlyn Graham** [3] **Anjay Friedman** [2] **Maximus Rafla** [2] **Coby Joseph** [2]

Abstract

Managing the risks of frontier AI will require international agreements.

Agreements require verification: an outside party must be able to check what a compute operator actually ran, for example constraining the total number of floating-point operations (FLOP).

Verification must be robust to an adversarial operator who hides computation, yet the hardware mechanisms proposed to catch one typically read silicon the operator controls.

Analogue side-channel measurements have been identified as a path forwards: sensors for power, electromagnetic, acoustic, or thermal emissions sit off the chip, beyond the operator's reach.

We quantify how tightly an off-chip power meter can constrain compute.

We derive a closed form for this tightness: the floor β, the largest covert computation a power trace cannot rule out, in units of one machine's declared capacity.

We measure every quantity entering the closed form on NVIDIA A100 GPUs.

We find the floor is large: we construct explicit cheating strategies that attain $\beta = 0.4$ behind a trace the meter accepts, and the closed form bounds the worst case at $\beta = 1.16$ when the declared number format can be checked, rising to $\beta = 1.95$ against a bare meter.

Additional verifier capabilities each buy a stated reduction — re-executing the declared work at an observed operating point brings β to 0.34 — but no capability reaches the covert side, which closes only by assumption on the adversary.

A power-based verification regime can then be designed — or rejected — on numbers rather than the intuition that joules imply operations.

[1]School of Mathematics and Statistics, The University of Melbourne, Parkville, VIC, Australia [2]RAND Corporation [3]Affiliation TBD. Correspondence to: Tom Kimpson <tom.kimpson@unimelb.edu.au>.

Preprint. July 26, 2026.

1. Introduction

As frontier AI systems grow more capable, governance proposals increasingly turn on *verification*: the ability of an outside party to check what a compute operator is doing (Reuel et al., 2025; Baker et al., 2025). One proposed mechanism is a threshold on compute (Sastry et al., 2024). Above a specified number of floating-point operations (FLOP), a training run triggers registration, reporting, or audit obligations (Shavit, 2023; Baker et al., 2025; Heim & Koessler, 2024). For example, U.S. Executive Order 14110 required reports on models trained using more than 10^{26} integer or floating-point operations (The White House, 2023), until its revocation in January 2025 (The White House, 2025).

A threshold in operation counts is only as good as the count it is applied to, so compute verification policies need to be robust to adversarial operators who do not declare honestly, but instead subvert the threshold by hiding computation. The mechanisms proposed to catch such an operator share a common weakness: they trust hardware or software that the operator controls. Utilisation counters, on-chip clocks, and memory telemetry all read silicon in the operator's hands, and so are conditional on tamper-resistant read paths and attested monitoring firmware that do not yet exist (O'Gara et al., 2025; Ansari, 2026; Rahman & Tajdari, 2026). Cryptographic proof-of-training protocols (Jia et al., 2021; Garg et al., 2023) need the prover's cooperation, and active challenge–response schemes (Monfared et al., 2026) must run their probes on the same untrusted hardware. An operator can even attempt to restructure its computation to evade compute-based oversight, though Seferis & Fist (2025) argue such structuring is itself detectable.

"Off-chip" analogue sensors have been suggested precisely to escape this trust dependence (Baker et al., 2025). Such sensors could be retrofitted to existing data centres and physically secured against tampering by the adversary, returning measurements of physical side channels—power, electromagnetic, acoustic, or thermal emissions—that the operator cannot edit in software.

There are reasons for optimism: in non-adversarial settings, physical emissions leak workload information. Emissions let an observer recover model architectures and hyperparameters (Hu et al., 2020; Maia et al., 2022; Gao et al., 2023; Chaudhuri et al., 2025), and software-visible counters and timing let a co-resident process do the same (Naghibijouybari et al., 2018; Wei et al., 2020). On-node telemetry and on-chip counters classify workloads (Tang et al., 2022; Xu et al., 2026; Gangwal et al., 2019), and power draw betrays anomalous computation (Clark et al., 2013) and running GPU activity (Arefin & Serwadda, 2024); a related body of work characterises power draw for GPUs and data centres (Patel et al., 2023; Choukse et al., 2025; Latif et al., 2025). But these are forward models, fitted largely in cooperative conditions: they map known workloads to their emissions. Verification poses the inverse problem—emissions to workload—against an operator actively trying to evade the sensor, and the question is how tightly analogue-sensor measurements can constrain compute in that setting.

That question is one of identification, not detection. Closest to our setting, Rahman & Tajdari (2026) evaluate the *detection* of hidden ML training from zero-overhead telemetry against an explicitly adversarial operator, across nine GPU models. Detection asks whether hidden work of a known kind trips a particular detector; we ask the complementary question—how much compute *any* verifier reading the sensor can bound, whatever detector it runs—and answer it with a closed-form floor rather than a classifier benchmark. A detector can raise alarms above the floor; it cannot narrow the identified set below the physics.

Of the physical side channels an off-chip sensor could read, we take power first: it already serves in the governance literature as a coarse proxy for compute, and it bears the most direct link to computation, since every operation costs energy and total energy therefore budgets total work. Whether the framework carries to the other channels is open, and we leave that to future work. Our contribution to the verification programme is the exchange rate between what the sensor reads and what the threshold counts.

This paper is organised as follows. Section 2 sets up the problem mathematically. We then derive a floor on the covert compute a power trace admits, in closed form from a handful of named physical bands (Section 3). Section 4 then asks how large that floor is: first from the physics of CMOS switching, then by measuring the bands directly on NVIDIA A100 hardware—one card for the band sweeps, three more for a cross-device check, together a stand-in for a full cluster—and assembling the floor a power meter allows (Section 4.3). Finally we walk that floor down a capability ladder (Section 5): each capability the verifier gains — a checked precision declaration, an observed clock, re-execution of the declared work, an assumption about the adversary's operands — moves one edge of the closed form, and the same measured bands price every rung the verifier can honestly price today. Section 6 sets out what transfers from device to facility scale, and what does not.

2. Setup and threat model

2.1. The forward model

Fix an observation window $t \in [0, T]$ and let $F(t)$ be the instantaneous rate of arithmetic operations [op/s], bounded by the hardware capacity $F_{\max}$. The forward model follows from the CMOS power law (Chandrakasan et al., 1992): chip power splits into a switching term and a static term,

$$P = \underbrace{\alpha C V^2 f}_{\text{dynamic}} + P_{\text{stat}}, \tag{1}$$

with α the activity factor, C the switched capacitance, V the supply voltage and f the clock. A datapath that retires N_{op} nominal operations per cycle runs at operation rate $F = N_{\text{op}} f$. Substituting $f = F/N_{\text{op}}$ and collecting constants gives the forward model

$$P(t) = r(t)\, F(t) + P_0(t), \qquad r \equiv \alpha \left(\frac{C}{N_{\text{op}}}\right) V^2, \tag{2}$$

where P_0 collects P_{stat} and the remaining non-compute overhead (idle draw, cooling, I/O). We call r the *exchange rate* [J/op]. Both equations hold instantaneously: α, C/N_{op} and V are set by whatever the datapath is executing at time t, and r inherits their time dependence. We carry the argument in $r(t)$, where that variation is the whole point, and suppress it in the definition. The clock has dropped out: what an operation costs does not depend on how fast the operations run, only on what each switching event costs and how many operations it delivers. Appendix A gives this derivation in full and locates the short-circuit and leakage terms.

The exchange rate $r(t)$ is not a constant of the device but a property of the execution, and it moves as the execution moves. Take its three factors in turn. The activity factor α depends on the operand values being multiplied (Chandrakasan et al., 1992). The capacitance switched per operation, C/N_{op}, depends on the numeric format—which sets both the datapath width and the packing N_{op}—and on where the operands live (Horowitz, 2014). The supply voltage V tracks the voltage–frequency operating point (Pallipadi & Starikovskiy, 2006). Every one of them is under the prover's control, and none is visible to an off-chip meter. Empirically the spread is wide: component-level analyses of a 45 nm process put arithmetic and on-chip memory access at 0.1–20 pJ per operation depending on format and operand locality (Horowitz, 2014), and our device-level measurements on a modern accelerator span 0.8–20 pJ/op (Section 4.3). More than an order of magnitude separates the cheapest operation a chip can retire from the dearest, and the prover chooses where in that range to sit.

The meter reports only a noisy version of this power,

$$P_{\text{obs}}(t) = P(t) + \eta(t), \tag{3}$$

with $\eta(t)$ a zero-mean measurement noise the prover cannot forge. The noise is a measurement limit: zero-mean noise with a suitable correlation structure (stationary, finitely correlated) averages away over the window and shrinks the longer the verifier watches. Throughout this paper we therefore assume a perfect meter and work with the clean power $P(t)$; every floor we derive is a floor on what even a noiseless meter can establish.

We count *hardware* operations—what the silicon actually executes—rather than model FLOPs, the algorithmically required operations, and we count them as *nominal operations* of whichever numeric format runs: a multiply–add retires two nominal ops whether it executes at fp32 or int8, and integer multiply–adds are ops, not FLOPs. Every executed operation costs joules, so the meter sees hardware operations.

2.2. The threat model

There are two parties. The *prover* is the monitored operator: it controls the workload, the operand values, the numeric formats, the clocks, and the idle states of the machine, and it may wish to run compute it does not declare. The *verifier* is the inspecting party: it reads P_{obs} from an off-chip meter it trusts, and it never sees r, F, or P_0 separately. The prover declares a workload of C_{dec} operations. The verifier wants to know if C_{dec} is consistent with P_{obs}. The verifier makes two assumptions:

Assumption 2.1 (Trusted meter)**.** The power trace cannot be forged: P_{obs} differs from the true draw P only by zero-mean noise, which we set to zero throughout.

Assumption 2.2 (Trusted declaration)**.** The declared computation really ran: the smallest total consistent with the window is C_{dec} itself. A power meter cannot by itself confirm this; independently checking that the declared work ran is a separate verification problem—the domain of proof-of-learning and zero-knowledge proof-of-training protocols (Jia et al., 2021; Garg et al., 2023).

Everything else—the exchange rate $r(t)$, the overhead $P_0(t)$, and how the operation rate $F(t)$ splits between declared and covert work—is under the prover's control and unobserved.

2.3. The inverse model

Define G as the forward map that carries a workload to its power trace,

$$G : \big(F(\cdot),\, r(\cdot),\, P_0(\cdot)\big) \;\longmapsto\; P(\cdot), \tag{4}$$

defined pointwise by Equation (2). The verifier faces an inverse problem: read the power trace and recover the workload that produced it, i.e. invert G. But G is *non-injective*. Since $r(\cdot)$ is free to vary, a low rate spent on much computation draws exactly the same power as a high rate spent on little. Distinct workloads (F, r, P_0) map to identical traces, and a many-to-one map has no inverse. Inversion therefore returns a set, not a point: the identified set $\mathcal{A}(P_{\text{obs}})$ collects every physically admissible workload that reproduces P_{obs} (Figure 1).

The question *"how many operations ran over the window?"* corresponds to one specific projection of $\mathcal{A}(P_{\text{obs}})$. Define the total compute

$$C \;=\; \int_0^T F\, dt \quad [\text{op}]. \tag{5}$$

Reading C off each admissible workload projects the identified set onto the C-axis,

$$\big\{\, C \;:\; (F, r, P_0) \in \mathcal{A}(P_{\text{obs}}) \,\big\} \;=\; [\, C_{\text{lo}},\, C_{\text{hi}} \,]. \tag{6}$$

We define β as the width of that interval, measured against the machine's peak declared-format capacity $C_{\max} = F^{\text{dec}}_{\max} T$,

$$\beta \;\equiv\; \frac{C_{\text{hi}} - C_{\text{lo}}}{C_{\max}}. \tag{7}$$

Under Assumption 2.2 the lower edge is the declaration itself, $C_{\text{lo}} = C_{\text{dec}}$, and the normaliser $C_{\max} = F^{\text{dec}}_{\max} T$ is the machine's capacity. The interval runs from that honest baseline up to the most compute the meter cannot rule out, so the upper edge is the declared work plus the largest covert load a cheater could slip past the meter,

$$C_{\text{hi}} \;=\; C_{\text{dec}} \;+\; C^{\max}_{\text{cov}}. \tag{8}$$

Substituting $C_{\text{lo}} = C_{\text{dec}}$ and Equation (8) into Equation (7) leaves a single unknown,

$$\beta \;=\; \frac{C_{\text{hi}} - C_{\text{lo}}}{C_{\max}} \;=\; \frac{C^{\max}_{\text{cov}}}{C_{\max}}. \tag{9}$$

The width $C_{\text{hi}} - C_{\text{dec}} = C^{\max}_{\text{cov}}$ is exactly the covert operations a cheater could hide. A wider interval means a larger β and more room for undeclared work; $\beta \gg 1$ means the ambiguity exceeds the entire machine.

Note that β is a floor, not a measurement error. It is the ambiguity that survives a perfect meter: because distinct workloads draw identical power, no reading of the trace can tell them apart because the data hold no information that separates them. Averaging, longer observation, and better estimators all fight measurement noise, and none of them lowers β.

3. Derivation of β

We now derive a closed form for β. The derivation is straightforward: integrate the forward model over the window, split the total energy into declared, covert, and overhead pieces, and ask how large the covert piece can grow

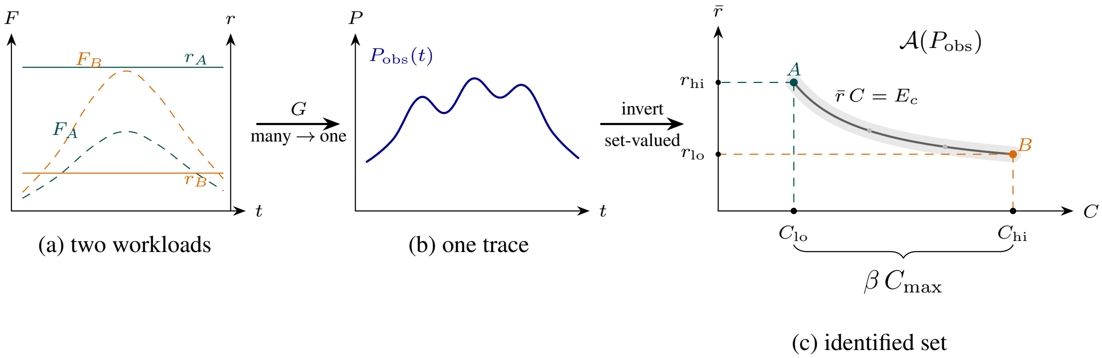


Figure 1. Non-injectivity of the forward map. (a) Two workloads (A little compute at a high rate, B much compute at a low rate) whose products rF coincide, so (b) they drive the identical power trace. (c) Inverting recovers only the set of consistent explanations $\mathcal{A}(P_{\text{obs}})$: the rate band $[r_{\text{lo}}, r_{\text{hi}}]$ cuts the constant-energy arc A–B, whose projection onto the C-axis spans $[C_{\text{lo}}, C_{\text{hi}}]$; relative to capacity that width is the floor β. The faint ribbon is meter noise (ignored in this work). Each rate is drawn constant only for clarity; in reality $r(t)$ varies in time. This does not matter for a question about total operations, which depends on $r(t)$ only through its op-weighted mean $\bar r = E_c/C$ (compute energy $E_c = \int_0^T rF\,dt$ over total operations C). That mean is panel (c)'s vertical axis, so letting $r(t)$ vary leaves the picture unchanged.

before the total stops looking honest. We work with the integrated trace (i.e. the total energy) rather than the resolved time series: treating power as affine in the operation rate (Equation (2)) lets its integral depend on the workload only through time-integrated quantities.

3.1. How much compute can hide?

Integrating the forward model Equation (2) over $[0, T]$, the left-hand side is the total energy the meter reports and the right-hand side falls into two pieces,

$$E \;=\; \underbrace{\int_0^T r(t)\,F(t)\,dt}_{\text{compute energy } E_c} \;+\; \underbrace{\int_0^T P_0(t)\,dt}_{\text{overhead } E_0}. \qquad (10)$$

The overhead E_0 is a nuisance the verifier knows only within a band $E_0 \in [E_{0,\text{lo}}, E_{0,\text{hi}}]$. We do not "subtract off" a known baseline: the prover controls fans, clocks, and idle states, so a known-and-removed P_0 would hand it a hiding channel.

The compute energy E_c splits into a declared block and a covert block,

$$\int_0^T rF\,dt \;=\; \underbrace{\int_0^T r^{\text{dec}}\,F_{\text{dec}}\,dt}_{\text{declared}} \;+\; \underbrace{\int_0^T r^{\text{cov}}\,F_{\text{cov}}\,dt}_{\text{covert}}. \qquad (11)$$

The declared block runs $C_{\text{dec}} = \int_0^T F_{\text{dec}}\,dt$ operations, taken known, and the covert block runs the excess $C_{\text{cov}} = C - C_{\text{dec}}$; this is the compute we are trying to bound. Write each block's energy as its operation count times a single mean cost per operation, i.e. the op-weighted averages

$$\bar r^{\,i} \equiv \frac{1}{C_i}\int_0^T r^i(t)\,F_i(t)\,dt \;=\; \int_0^T r^i(t)\,\frac{F_i(t)}{C_i}\,dt, \qquad (12)$$

for $i \in \{\text{dec}, \text{cov}\}$. The second form makes the averaging explicit: the weight $F_i(t)\,dt/C_i$ is the fraction of the block's operations that run in $[t, t+dt]$; it is nonnegative and integrates to one. The machine therefore draws

$$E \;=\; \underbrace{\bar r^{\text{dec}}\,C_{\text{dec}}}_{\text{declared work}} \;+\; \underbrace{\bar r^{\text{cov}}\,C_{\text{cov}}}_{\text{covert work}} \;+\; \underbrace{E_0}_{\text{overhead}}, \qquad (13)$$

and the verifier sees only E.

Both blocks run on the same silicon, so both mean costs are bounded by the *same* physical envelope $[r_{\text{lo}}, r_{\text{hi}}]$, the hardware's full range of energy per operation. We nonetheless track them on separate labels, $\bar r^{\text{dec}} \in [r^{\text{dec}}_{\text{lo}}, r^{\text{dec}}_{\text{hi}}]$ and $\bar r^{\text{cov}} \in [r^{\text{cov}}_{\text{lo}}, r^{\text{cov}}_{\text{hi}}]$, because a verifier with channels beyond the meter may come to know more about one than the other; with only a power meter the two coincide with that common envelope.

The verifier raises no alarm so long as the measured total is explainable by honest declared work at some admissible rate and overhead. The most an honest run could ever draw is the declared work at the top of its band, on top of the highest overhead,

$$E \;\le\; r^{\text{dec}}_{\text{hi}}\,C_{\text{dec}} \;+\; E_{0,\text{hi}}. \qquad (14)$$

The test is deliberately one-sided: covert work can only *add* energy, so a draw is suspicious only when the total runs *high*. A symmetric lower alarm would flag *under*-delivery—drawing too little to be the declared work—which is outside the covert-compute threat model, so no lower bound is imposed (Theorem B.1(ii)). Substituting the budget Equation (13) into the no-alarm condition Equation (14) and rearranging,

$$C_{\text{cov}} \;\le\; \frac{\left(r^{\text{dec}}_{\text{hi}} - \bar r^{\text{dec}}\right) C_{\text{dec}} \;+\; \left(E_{0,\text{hi}} - E_0\right)}{\bar r^{\text{cov}}}. \qquad (15)$$

A cheater now pushes every free choice on the right-hand side to its limit:

- *Run the declared work cheaply,* $\bar{r}^{\text{dec}} \to r_{\text{lo}}^{\text{dec}}$. The verifier cannot tell an expensive declared run from a cheap one, so it must allow the draw all the way up to $r_{\text{hi}}^{\text{dec}}\, C_{\text{dec}}$; the gap $(r_{\text{hi}}^{\text{dec}} - r_{\text{lo}}^{\text{dec}})\, C_{\text{dec}}$ is free energy the prover spends covertly while the total still reads as honest work.
- *Idle the overhead,* $E_0 \to E_{0,\text{lo}}$, freeing the full band width $\Delta E_0 = E_{0,\text{hi}} - E_{0,\text{lo}}$.
- *Run the covert work cheaply,* $\bar{r}^{\text{cov}} \to r_{\text{lo}}^{\text{cov}}$, so each hidden operation costs as little as the hardware allows.

At those values Equation (15) attains its maximum, the largest covert load the *energy budget* admits,

$$C_{\text{cov}}^{\text{max, en}} \;=\; \frac{\left(r_{\text{hi}}^{\text{dec}} - r_{\text{lo}}^{\text{dec}}\right) C_{\text{dec}} \;+\; \Delta E_0}{r_{\text{lo}}^{\text{cov}}}. \tag{16}$$

Divide by $C_{\max}$ and write the declared *hardware utilisation* $h = C_{\text{dec}}/C_{\max}$ for the fraction of the machine's declared-format capacity the declared work occupies. What results is the identified-set width of Equation (7) with the admissible set cut by the energy constraint alone — the same normalised width as β, computed as though the no-alarm condition Equation (14) were the only thing standing between the prover and unlimited covert work. We write it β_{en}, the *energy-only width*,

$$\beta_{\text{en}}(h) \;=\; h\,\frac{r_{\text{hi}}^{\text{dec}} - r_{\text{lo}}^{\text{dec}}}{r_{\text{lo}}^{\text{cov}}} \;+\; \frac{\Delta E_0}{r_{\text{lo}}^{\text{cov}}\, C_{\max}}\,. \tag{17}$$

It is a component of the floor, not yet the floor itself: energy is one constraint on covert work, and Section 3.2 adds the other. Pedagogically, Equation (17) describes (how much honest work there is) $\times$ (how uncertain its per-operation cost is), divided by (how cheaply covert work can run), plus a smaller slack from the overhead band. The first term grows with the declared-band width and with the declared utilisation: more legitimate work offers more cover. The second term is the overhead's contribution: joules the prover frees by idling overhead that the verifier must still allow for.

Note the asymmetry between the numerator and denominator of Equation (17). Declarations and re-execution can narrow the declared band width $r_{\text{hi}}^{\text{dec}} - r_{\text{lo}}^{\text{dec}}$ in the numerator, but no declaration reaches the covert floor $r_{\text{lo}}^{\text{cov}}$ in the denominator; only hardware-wide observables (clock, voltage) can raise $r_{\text{lo}}^{\text{cov}}$.

The two edges therefore play different roles, and must be measured differently: the declared band width tolerates a total-quotient estimate, while the covert denominator admits only a marginal one. Appendix C states the condition precisely, and it is weaker than the affine map of Equation (2).

3.2. The capacity clip

The energy budget is not the only physical constraint: covert work must also *fit on the machine*. The setup already bounds the operation rate by the hardware capacity (Section 2.1), and that capacity is format-dependent. The peak rate $F_{\max}$ is not one number but a property of the executing format: an A100's dense peaks run from 19.5 Top/s at fp32 on the CUDA cores through 156 at tf32 and 312 at fp16/bf16 to 624 Top/s at int8 on the tensor cores (NVIDIA Corporation, 2020). We write $F_{\max}^{\text{dec}}$ for the peak of the declared format, which fixes the capacity normaliser $C_{\max} = F_{\max}^{\text{dec}}\, T$ used throughout, and

$$\gamma \;=\; \frac{F_{\max}^{\text{cov}}}{F_{\max}^{\text{dec}}} \tag{18}$$

for the capacity ratio of a covert format against the declared one ($\gamma = 2$ for int8 covert work against fp16 declared work on an A100).

However much energy slack the trace leaves, then, the covert block can never retire more than the capacity the declared work leaves free. Declared work occupies a fraction h of the declared-format capacity; the remainder, executed at the covert format's peak, leaves a second width, in the same units as β_{en}:

$$C_{\text{cov}}/C_{\max} \;\leq\; (1-h)\,\gamma. \tag{19}$$

We evaluate Equation (19) at the *published peak* capacities rather than the rates our kernels achieve: peaks give the covert stream more room than it can really use, so this is the adversary-favorable—and for an upper bound on the ambiguity, conservative—direction (Appendix D quantifies the difference on our measured peaks). Each width is what Equation (7) returns when one constraint is imposed and the other ignored, so the floor—the width when both hold at once—is the smaller of them,

$$\beta(h) \;=\; \min\!\big[\beta_{\text{en}}(h),\; (1-h)\,\gamma\big]. \tag{20}$$

The two pull in opposite directions. Energy *rises* with h—more declared work offers more cover—while capacity *falls*—more declared work leaves less machine to hide on. At full declared utilisation the machine is spoken for and $\beta(1) = 0$; at idle there is capacity but almost no cover. The worst case sits at the crossing,

$$h^* = \frac{\gamma - b}{s+\gamma}, \qquad \beta^* = \gamma\,\frac{s+b}{s+\gamma}, \tag{21}$$

writing $\beta_{\text{en}}(h) = s\,h + b$ for its slope and intercept. The crossing form holds when $b < \gamma$; if the overhead term alone exceeds the clip ($b \geq \gamma$) the clip binds everywhere and the maximum sits at the boundary, $(h^*, \beta^*) = (0, \gamma)$, as the released code implements. Each side of the minimum is the binding constraint in its own regime: energy binds for $h < h^*$ (the trace itself caps the covert load), capacity binds for $h > h^*$, and both bind at the crossing. Where the crossing sits is an empirical question about s. On jointly consistent bands—one declared format's own band over the covert format's marginal cost—the measured slope is of order one (Section 4.3), so the crossing falls at intermediate utilisation and the worst case is a fraction of γ. We still quote β_{en} on its own as a diagnostic— $\beta_{\mathrm{en}} > 1$ says the energy budget alone cannot rule out a machine's worth of covert work—but only β is the width of the identified set.

Format bookkeeping. Format dependence makes joint feasibility a modelling obligation, not a detail. Every floor we evaluate for a verifier that has *pinned the declared format*—the headline and every rung below it—is computed inside a single jointly feasible scenario: one declared format and one covert format sharing one observation window. The declared band, the capacity normaliser $C_{\max}$, and the utilisation h all refer to the same declared format; the covert cost and the covert peak to the same covert format; and the two streams time-multiplex the machine, so covert work holds at most the $(1-h)$ share of resource-time the declared work leaves free. We allow the prover to switch formats between the two streams at zero cost—the adversary-favorable direction, since any real switching overhead only shrinks the covert share. What the model forbids is combining extrema that no single schedule attains: an honest interpretation priced at one format's dearest cells is only admissible if that format can execute the declared work at rate $h\,F^{\mathrm{dec}}_{\max}$ at all, and the headline path in our analysis code rejects any cross-format combination of extrema by construction (Section 4.3).

Within one such scenario the bands enter as a rectangle—each edge attainable independently of the others—and Equation (20) prices the cheating corner of that rectangle. Whether any single schedule sits at that corner is the one substantive assumption behind the floor.

Assumption 3.1 (Joint feasibility of band corners)**.** There exist admissible workloads that attain the cheating corner $(\bar r^{\mathrm{dec}}, E_0, \bar r^{\mathrm{cov}}) = (r^{\mathrm{dec}}_{\mathrm{lo}}, E_{0,\mathrm{lo}}, r^{\mathrm{cov}}_{\mathrm{lo}})$ jointly, and under shared constraints: the *same* observation window T, the *same* trusted declared work C_{dec}, one device in a *common thermal and clock state* whose resource-time the declared and covert streams multiplex at their *achieved* (not nominal-peak) throughputs, and the per-format capacities of Equation (19).

Without Assumption 3.1 the rectangle is an *outer relaxation* of the true feasible set, and every width in this section is an upper bound rather than an exact width. Our measurements probe the assumption from below rather than establish it: the matched-energy witnesses of Section 4.3 exhibit workloads meeting these shared constraints whose sensor-accepted cheating points reach $h \approx 0.22$, but their achieved rates do not jointly reproduce the rectangle's extreme edges, so they attain *lower witnesses* for the width (0.41 machines at $h = 0.216$, against the model's 0.72), not the corner itself. The corner remains an assumption at every h, so every width we quote is an upper bound, met from below by the witnesses.

3.3. The power-only floor

A power meter reports total energy and nothing more: to it an operation is an operation, declared or covert, so there is no way to constrain the declared rate separately from the covert one. The two bands have no choice but to coincide, both equal to the full measured envelope (five tested formats; the hardware's true envelope, holding unmeasured int4 and binary paths, is wider still): $r^{\mathrm{dec}}_{\mathrm{hi}} = r_{\mathrm{hi}}$ and $r^{\mathrm{dec}}_{\mathrm{lo}} = r^{\mathrm{cov}}_{\mathrm{lo}} = r_{\mathrm{lo}}$. Setting these equal in Equation (17) collapses the energy-only width to its single-band, power-only form

$$\beta^{\varnothing}_{\mathrm{en}}(h) \;=\; h\left(\frac{r_{\mathrm{hi}}}{r_{\mathrm{lo}}} - 1\right) \;+\; \frac{\Delta E_0}{r_{\mathrm{lo}}\,C_{\max}}, \tag{22}$$

where the subscript $\varnothing$ denotes the empty set of extra channels: the meter is all the verifier has. This rung is the deliberate exception to the single-scenario discipline of Section 3.2: a bare meter pins no format at all, so both band edges open to the cross-format envelope while $C_{\max}$ and h stay the declared-format yardstick, and we flag the rung as a cross-format envelope by intent wherever it is quoted.[1] The capacity clip must be re-derived here too, because with no format pinned on the *declared* side either, the cheating schedule executes the declared operations themselves in the cheap covert format: they occupy only h/γ of the machine's resource-time rather than h, declared plus covert work together obey $C_{\mathrm{cov}} \le (\gamma - h)\,C_{\max}$, and the power-only floor is $\beta(h) = \min[\,\beta^{\varnothing}_{\mathrm{en}}(h),\, \gamma - h\,]$.[2] Two limits bracket the first term of Equation (22). First is *ambiguity*: if the rate is known exactly ($r_{\mathrm{hi}} = r_{\mathrm{lo}}$) the term vanishes, since a pinned energy budget converts covert joules straight back into operations. Second is *opportunity*: if the machine is idle ($h = 0$) it vanishes too, with no declared work to hide beneath. The overhead term, by contrast, needs no declared cover at all.

A worked example. To put a number on the power-only floor, take $C_{\max} = 100$ operation-units, declared work $C_{\mathrm{dec}} = 50$, a rate known only to lie in $[1.5, 2.5]$, and an overhead band $E_0 \in [10, 20]$. A cheater runs $C = 90$

[1] The combination is still admissible under the single-schedule criterion of Section 3.2: the honest acceptance ceiling is the fp32 rate, and fp32 can sustain the declared rate it implies ($h^* C_{\max} = 14.3$ Top/s against a measured 17.7 Top/s fp32 peak).

[2] A slightly laxer clip than the single-declared-format $(1-h)\,\gamma$ of Equation (19), which would charge the declared work the covert format's energy but the declared format's time share; the two coincide at $\gamma = 1$.

at the efficient end $\bar{r} = 1.5$ with the overhead idled to $E_0 = 10$, drawing $E = 1.5 \times 90 + 10 = 145$: exactly the energy of honest declared work at the expensive end under high overhead, $2.5 \times 50 + 20 = 145$. No alarm. The verifier cannot separate the honest 50 from the true 90—not for want of statistical power but structurally: the two hypotheses are energy-degenerate, and no test on the total can separate them. Of the 40 covert units, $\approx$33 hid under the rate ambiguity of the declared work (the first term of Equation (22)) and $\approx$7 under the overhead band (the second). In the language of Equation (7): $C_{\rm lo} = 50$, $C_{\rm hi} = 90$, so $\beta = 40/100 = 0.4$. Equation (22) assembles the same number term by term: $\beta^{\varnothing}_{\rm en}(0.5) = 0.5\,(2.5/1.5 - 1) + 10/(1.5 \times 100) = 1/3 + 1/15 = 0.4$. The cheat is also capacity-feasible—the 40 covert units fit inside the $(1-h)\,\gamma\,C_{\max} = 50$ units the declared work leaves free (taking $\gamma = 1$)—so here the clip of Equation (19) is slack and $\beta(0.5) = 0.4$. Had the slack exceeded 50 units, the identified set would have ended at the capacity edge instead: at full declared utilisation β vanishes no matter how wide the rate band is.

What β is, and is not. Two readings of the width must be kept apart. Given an *observed* total E, the identified set has a width that depends on how much energy was actually drawn. β is instead the largest such width over every total the no-alarm rule accepts: a design-time statement about what *any* accepted window could hide, not the ambiguity of the window in front of you. Quoting the second as though it were the first overstates a typical audit, and quoting the energy-only width $\beta_{\rm en}$ without its capacity clip overstates what the silicon admits. Appendix B separates the two readings, together with the one-sidedness of the acceptance rule, as four formally distinct estimands.

4. How big is β?

So far the floor is only a pair of formulas—the energy-only width Equation (22) and its capacity clip Equation (19); this section asks how large they actually are. We first analyse the slack's two terms and their physical drivers (Section 4.1), then pin down the value two ways: an order-of-magnitude estimate from textbook physics (Section 4.2), and a direct measurement on an NVIDIA A100 (Section 4.3).

4.1. The two terms

The energy-only width Equation (22) is a sum of two terms: an ambiguity term $h\,(r_{\rm hi}/r_{\rm lo} - 1)$, set by the spread of the exchange rate, and an overhead term $\Delta E_0/(r_{\rm lo}\,C_{\max})$, set by the width of the overhead band. We take each in turn.

The exchange-rate term. Its size is governed by the ratio $r_{\rm hi}/r_{\rm lo}$. A chip's power divides into a dynamic (switching) part and a static part; the exchange rate comes from the dynamic part, the CMOS switching power $P_{\rm dyn} = \alpha C V^2 f$ (Chandrakasan et al., 1992), whose cost per operation factors into

$$r \;\approx\; \kappa(\text{precision}) \cdot \alpha(\text{operands}) \cdot C_{\rm eff}(\text{locality, kernel}) \cdot V^2. \tag{23}$$

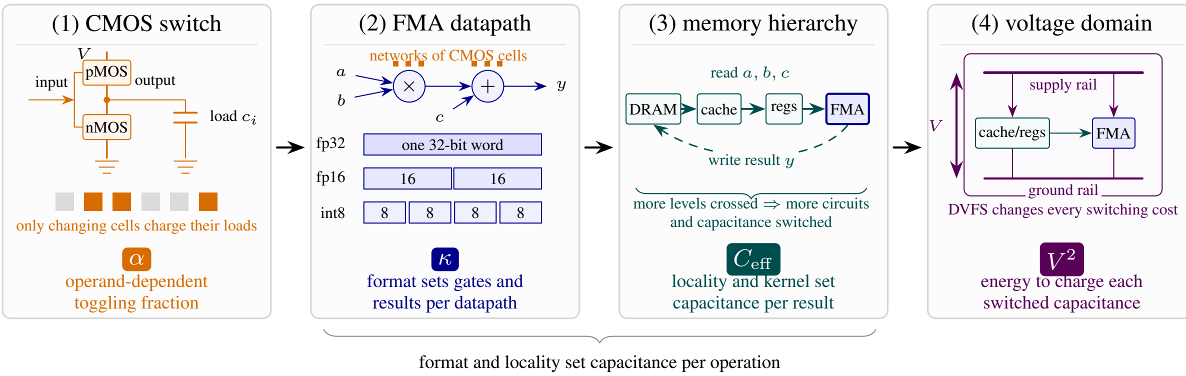
Appendix A derives this factorisation; the static part—leakage, cooling, and idle draw—does not enter r but resurfaces as the overhead term below. Each factor is set by a different physical scale of the machine (Figure 2). The numeric-format scale κ falls with datapath width at each precision step (fp32 $\to$ int8), tracking the pJ-scale energy hierarchy of arithmetic (Horowitz, 2014); the activity factor α is the data-dependent fraction of transistors that toggle for the given operands; the effective capacitance $C_{\rm eff}$ depends on where operands live and which kernel runs. Less-local operands activate more of the memory hierarchy—storage arrays and peripheral logic, buffers and interfaces, as well as interconnect—so their combined dynamic cost is represented by a larger effective switched capacitance (Horowitz, 2014; Chatterjee et al., 2017); and V^2 is the capacitor-charging energy of the supply voltage, which DVFS slides along the voltage–frequency curve (Pallipadi & Starikovskiy, 2006). With the power meter as the only channel, every one of these factors is unknown to the verifier and conceded at once.

The overhead term. The overhead P_0—idle draw, cooling, leakage, and I/O—admits no comparably clean factorisation. The exchange rate, though it too varies with the hardware and its operating point, obeys a single switching law that resolves into a few factors with bounded ranges; the overhead obeys no such law. It is a heterogeneous mix of static leakage, cooling control loops, firmware idle states, and platform I/O, so no single relation fixes the band width ΔE_0 from first principles.

4.2. Back of the envelope

Taking the factors in order: the precision scale κ spans roughly an order of magnitude from fp32 down to int8 (Horowitz, 2014) (our own sweep in Section 4.3 finds 13.7× from fp32 to int8 at dense operands); the operand activity α adds a data-dependent factor of order two (no textbook number exists; we measure it directly in Section 4.3); locality and kernel set $C_{\rm eff}$, for which we conservatively count no extra factor;[3] and DVFS moves V^2 by a factor of about

[3] Per access, DRAM costs about a hundred times more than on-chip memory (Horowitz, 2014), but the per-operation effect is diluted by the kernel's bytes-per-operation: negligible for a compute-bound matmul (as Section 4.3 finds), though potentially large for a memory-bound kernel.



$$r = \kappa \cdot \alpha \cdot C_{\text{eff}} \cdot V^2$$

f cancels from the marginal rate; leakage and cooling leave as the overhead P_0, not r

Figure 2. **The exchange rate, from transistors to the powered die.** Each physical scale contributes one factor to the per-operation energy of Equation (23). (1) A complementary pMOS–nMOS inverter charges or discharges its effective load c_i when its input changes; an operand determines what fraction α of all such cells toggle. (2) Networks of these cells implement the multiply and add of an FMA. The numeric format selects the word width (and supported packing), changing the logic engaged per arithmetic result and hence κ. (3) The FMA reads operands through the register–cache–DRAM hierarchy and writes its result back. Traversing more of that hierarchy switches additional array and peripheral circuits, buffers and interfaces, and interconnect; their combined dynamic cost is represented by C_{eff}. (4) Within a voltage domain, the supply places V across the arithmetic and interconnect loads, so DVFS contributes V^2 per unit capacitance. The clock f cancels from the marginal rate (Equation (28)) and the static draw—leakage, cooling—leaves as the overhead P_0, not r. Multiplying the factors recovers Equation (23).

1.5.[4] Multiplied, the full hardware ratio $r_{\text{hi}}/r_{\text{lo}}$ sits conservatively in the tens—but that span crosses numeric formats, and no single declared workload can be honestly explained at one format's dearest cells while its capacity is priced at another's (Section 3.2). Within one declared format the precision factor drops out, leaving operands and DVFS: a jointly feasible declared band of a factor 2–3, so the slack slope s is of order one and the ambiguity term reaches, not dwarfs, one machine. The overhead term, which we cannot fix from first principles (Section 4.1), is additive and only raises the slack. On paper, then, energy binds at low utilisation, the capacity clip of Equation (19) takes over at high utilisation, and the worst-case floor sits at a substantial fraction of the machine—up to about its declared-format capacity as a model bound. A bare power meter certifies little about how many operations ran. We put numbers on this next by direct measurement.

4.3. Measurement on an NVIDIA A100

We now turn the CMOS factorisation of Equation (23) into a measurement plan. Its factors do not enter Equation (22) separately. Dividing dynamic switching power by operation rate cancels the clock frequency and leaves κ, α, C_{eff}, and V^2 to determine the dynamic energy per operation r; the range they allow is summarised by the exchange-rate edges r_{hi} and r_{lo}. Experiment 1 sweeps precision (κ) and locality (C_{eff}) to find the observed ceiling; Experiment 2 sweeps operand activity (α) and takes its cheapest point as the observed floor. The platform did not expose voltage control, so we could not sweep V^2; we record the governor-selected clock as operating-point metadata instead. The static component of the power omitted by Equation (23) returns as the intercept P_0 of the affine power law; Experiment 3 measures the width of that intercept band and hence ΔE_0 over the observation window. The three experiments therefore collapse the transistor-level picture into exactly the three macroscopic inputs required by Equation (22): r_{hi}, r_{lo}, and ΔE_0.

Setup and provenance. Experiments 1–4 ran sequentially on one NVIDIA A100-SXM4-80GB in a single Slurm allocation; a compact cross-device sweep on three more cards of the same SKU gives the 2–3% device-to-device spread reported with the matched-energy witnesses below, and the useful-operand workloads (Experiment 5) and matched-energy witnesses ran on further same-SKU cards. The platform denied clock locking and power capping, so DVFS is a recorded axis rather than a controlled one: each window logs the governor-selected SM clock and the bands are stratified into clock strata. Energy is read from the on-die NVML total-energy counter; each measured cell fills a $\sim$1.5 s window with back-to-back kernels on pre-allocated buffers and keeps the per-cell median over repeated windows. This is a non-adversarial, device-level calibration: the telemetry

[4] Supply voltages on modern accelerators sit near 1 V (Horowitz, 2014) and DVFS moves them by roughly $\pm$10%; energy per operation scales as the ratio squared, $(1.1/0.9)^2 \approx 1.5$.

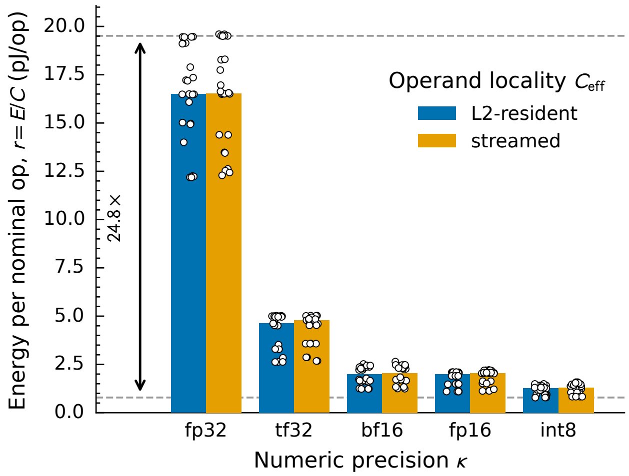


Figure 3. The exchange-rate band of Equation (22) (Experiment 1). Total energy quotient E/C for a fixed matmul kernel on an NVIDIA A100, swept factorially over precision κ, operand locality $C_{\rm eff}$ (paired bars), and operand distribution α; white points are the individual measured windows ($n = 240$ across 60 cells). Dashed lines mark the band edges: the dearest cell is the band ceiling $r_{\rm hi}$, the cheapest (int8, zero operands) the envelope floor $r_{\rm lo}$.

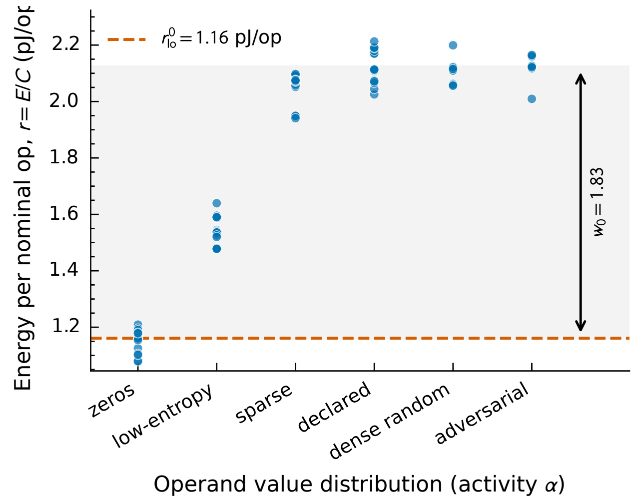


Figure 4. The operand envelope (Experiment 2, $n = 96$ windows across six operand distributions). Kernel, shape, and precision (fp16) are held fixed and only the operand values vary; the clock was governor-selected, not locked, so the map is descriptive (see text). The shaded span is the observed operand spread $w_0 = 1.83$; the minimum (zeros, dashed) is the descriptive fp16 operand floor $r^0_{\rm lo} = 1.16$ pJ/op as a total quotient—*not* the covert denominator, which is Experiment 4's marginal edge.

is trusted as a laboratory instrument, not as the verifier's sensor in the adversarial setting of Section 2, and one GPU suffices to estimate the bands and the order of magnitude of β; extending the calibration to off-chip meters and multi-device systems is natural future work. Appendix F gives the full protocol, the meter tolerance, and the provenance manifest, including which card measured which band.

Estimands. The experiments estimate the two quantities of Appendix C, each where it is valid. Experiments 1 and 2 report per-window *total quotients* E/C at sustained, fully busy rates: the right estimand for the declared band, which enters the floor only through the difference of its edges, where the amortised baseline cancels (Appendix F). The covert denominator needs the opposite, one-sided guarantee—a lower bound on the *incremental* cost of an added covert operation, for which a total quotient is anti-conservative—so it is identified by Experiment 4's dose–response intervention and used at its one-sided lower confidence limit. Experiment 3's between-shape regression of power on rate is a consistency diagnostic for the affine model, never a theorem input; its idle cells pin the intercept band ΔE_0.

Experiment 1—the exchange-rate envelope. A compute-bound matmul (operation count $C = 2MNK$ known analytically; the int8 cells retire integer multiply–adds, counted as nominal operations per the convention of Section 2.1), swept factorially over precision (κ, fp32 down to int8), operand locality ($C_{\rm eff}$, operands resident in vs. streamed through the L2 at a byte-identical shape), and operand distribution (α, zeros through adversarial-toggle), stratified by the governor-selected clock. The pooled envelope spans $r_{\rm lo} = 0.786$ pJ/op (int8, zero operands) to $r_{\rm hi} = 19.5$ pJ/op (fp32), a span of 24.8× (Figure 3), with precision following the textbook ordering int8 $<$ bf16 $\approx$ fp16 $<$ tf32 $<$ fp32. Precision drives most of the spread; operand values contribute a further factor ≈2 within each format, and locality moves r little for this compute-bound kernel. The pooled span is descriptive of the range the five tested formats span (the hardware's unmeasured int4 and binary paths sit outside it); the floor never reads it, because its two edges live in different formats. The jointly feasible declared band for the fp16 case study is the fp16 cells' own band, $[1.10,\ 2.19]$ pJ/op—a span of 2.0×, set by operands and DVFS with the precision factor held fixed.

Experiment 2—the operand-driven spread of r. The same matmul at fixed kernel, shape, and precision (fp16), now varying only the operand *values* (zeros, structured sparsity, low-entropy, declared-typical, dense-random, adversarial maximum-toggle). The SM clock, however, was governor-selected rather than locked (locking is privilege-denied; Section 4.3), and the operand families did not run at a common clock—the cheapest (zeros, low-entropy) drew the highest clocks and the dearest (dense-random, adversarial-toggle) the lowest. This experiment therefore does not cleanly isolate the activity factor α; we read it as descriptive within-device evidence of the operand-driven

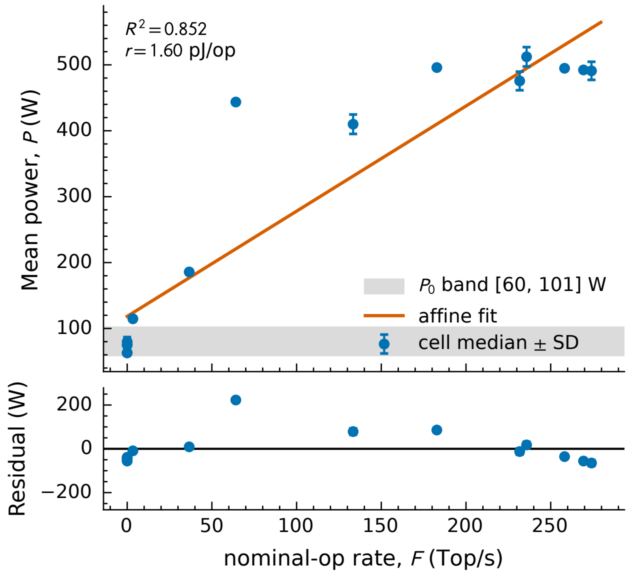


Figure 5. The overhead band and the affine forward map (Experiment 3, $n = 65$ windows across a 15-cell rate ladder). *Top:* mean power P vs. nominal-op rate F, plotted as per-cell medians with ± 1 SD error bars over each cell's repeats, against the affine fit $P = rF + P_0$ ($R^2 = 0.852$; slope $r = 1.60$ pJ/op); the shaded band is the observed idle range across five idle variants, $P_0 \in [59.8, 101.0]$ W. *Bottom:* residuals; the launch- and bandwidth-bound mid-ladder shapes carry the scatter. This is a *between-shape* diagnostic of the affine map; the covert denominator the floor actually uses is the *within-cell* marginal cost of Figure 6 (Experiment 4), not this fit.

spread of r, not as a certified population edge. Operand values alone spread r by an observed factor $w_0 = 1.83$ (Figure 4), following the ordering zeros $<$ low-entropy $<$ structured-sparse $<$ dense-random $\approx$ adversarial-toggle (Appendix G constructs the distributions explicitly). The clock confound plausibly *compresses* this spread rather than inflating it—the cheap families ran fastest, so a common clock would, if anything, spread the energies further—but we offer this only as a direction-of-bias note, not a proof: DVFS voltage scaling, throughput, and baseline-power amortisation pull in competing directions, and one-sided uncertainty cannot repair an uncontrolled confound. No security bound rests on w_0 or on this operand edge. The minimum-energy run (zeros) puts the fp16 operand floor at 1.16 pJ/op as a total quotient—a descriptive edge only; the covert denominator the floor actually uses is Experiment 4's marginal cost (a one-sided 95% lower confidence limit), which strips the baseline power this quotient amortises in.

Experiment 3—the overhead band ΔE_0 and the marginal rate. The *rate ladder* is ten fp16 matmul shapes chosen to span the machine's sustained throughput: three small cubes (256^3 to 1024^3) that are launch-bound, two skinny-K shapes ($8192 \times 8192 \times 64$ and $\times 256$) that are bandwidth-bound, and five large cubes (2048^3 to 8192^3) that are compute-bound—a 60–90$\times$ spread in sustained operation rate at a single numeric format. Each window runs 100% busy at its shape's own natural rate. We deliberately do *not* vary F by duty-cycling one kernel against idle: averaging a fixed busy power against a fixed idle power is linear in the duty fraction by construction, so that design would manufacture the affine form rather than test it. Operation counts are analytic ($C = 2MNK$ per matmul, Section 2.1), so the abscissa is an exact op count divided by the measured wall time of the window and the ordinate is metered mean power. Five idle variants—before any CUDA context, context up, context plus operand pool resident, immediately after the hottest ladder cell, and after a 60 s cooldown—supply the $F = 0$ anchors.

The idle range, widened by the adopted $\pm 5\%$ meter tolerance, pins the overhead band to $P_0 \in [59.8, 101.0]$ W (width $\Delta P_0 = 41.2$ W). This band is read directly off those five idle cells, not from a fitted intercept; no input to the floor comes from the fitted slope, which serves only as a diagnostic of the forward model.

As that diagnostic, the affine map holds tightly within a kernel class and loosens across classes. Restricted to the idle anchors and the five compute-bound cubes—like compared with like—the fit gives 1.65 pJ/op at $R^2 = 0.99$. Opened to all ten shapes it gives 1.60 pJ/op at $R^2 = 0.852$ (Figure 5): the slope moves by 4% while the scatter grows sharply, so the launch- and bandwidth-bound shapes account for the residual and not for the rate. This is the expected behaviour rather than a failure of the model. Those shapes move C_{eff} and α along with F, and the resulting spread in r is the same physical variation Experiment 1 maps deliberately as the exchange-rate band—swept on purpose there, drifting as a side effect here. The between-shape slope is a diagnostic of the affine forward model, not the floor's covert denominator: it averages over shapes whose non-covert factors vary together, where the theorem needs the derivative at fixed declared work. That derivative is what Experiment 4 identifies.

Experiment 4—the marginal covert cost. The covert denominator, measured as the theorem defines it: the energy an *added* covert operation costs, with everything else held fixed. Every measured window has the same fixed 10 s wall length and contains the same calibrated fp16 declared burst (utilisation 0.30 of the window, declared-typical operands); to it we add N_{cov} covert matmuls of one format (int8 or fp16, zero operands—the adversary's cheapest case), then idle-pad to the window. Sweeping the covert dose over six levels from zero to a 0.48 busy fraction (randomised order, drift-reference splices, five repeats per cell) and regressing window energy on covert nominal operations gives the marginal cost as a slope (Figure 6): 0.53 pJ/op for int8 and 0.77 pJ/op

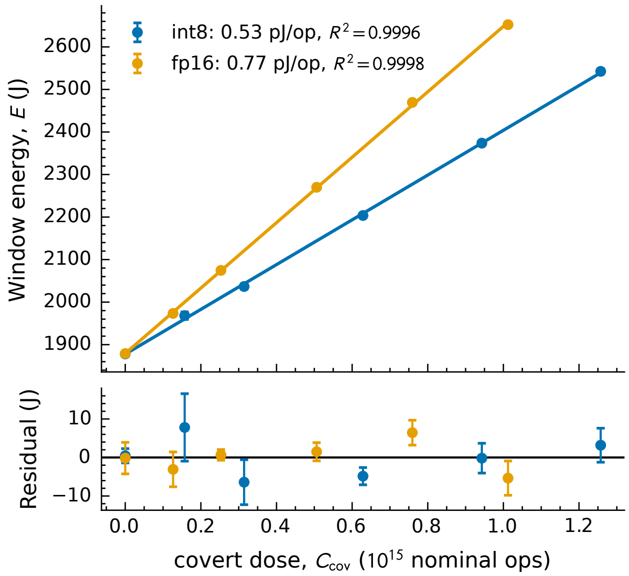


Figure 6. The marginal covert cost (Experiment 4): window energy E vs. covert dose C_{cov} for int8 and fp16 covert work, added to a fixed fp16 declared burst inside a fixed 10 s window. *Top:* per-cell medians with ± 1 SD error bars over each cell's four untrimmed repeats (of five captured; the flagged high-tail repeat is excluded), against the dose fit $E = r_{\mathrm{cov}}\, C_{\mathrm{cov}} + E_0$; both slopes are linear to $R^2 > 0.999$ (0.53 pJ/op int8, 0.77 pJ/op fp16) and share a zero-dose intercept matching the directly measured zero-dose windows to 0.02%. *Bottom:* residuals. Because the baseline burns identically at every dose, it cancels in the slope: unlike the between-shape fit of Figure 5, this within-cell slope is the incremental estimand $\partial E/\partial C_{\mathrm{cov}}$ the floor's covert denominator requires (Appendix C).

for fp16, both fits linear to $R^2 > 0.999$, with the fitted zero-dose intercept agreeing with the directly measured zero-dose windows to 0.02%. Because the fixed window burns the same baseline energy at every dose, baseline power cancels in the slope by construction—the marginal cost sits 0.26 (int8) to 0.39 pJ/op (fp16) below the corresponding total quotients, which amortise it in. The floor uses the one-sided 95% *lower* confidence limits, $r^{\mathrm{cov}}_{\mathrm{lo}} = 0.51$ pJ/op (int8) and 0.76 pJ/op (fp16): the covert cost divides the slack, so underpricing covert work is the security-safe direction.[5] Clock locking is denied on this platform, so the governor is a recorded, not controlled, axis here too; the window-median clock varies with the dose through the idle-pad share, and the identification rests on the dose-response linearity and the intercept anchor rather than on a common clock stratum—a stated limitation of the platform.

[5]One-sided in the statistical sense only: the limit controls sampling uncertainty in the fitted dose slope for this workload family, device, and driver state. It is not a uniform physical lower bound over every execution the threat model admits—unseen kernels, schedules, operand encodings, or thermal states could in principle sit below it. Every rung that divides by $r^{\mathrm{cov}}_{\mathrm{lo}}$ is therefore conditional on the measured covert families bracketing the adversary's cheapest edge; the same reading applies to the useful band of Experiment 5, estimated from one trained transformer layer at two shapes.

Experiment 5—the marginal *useful* covert cost. The same intervention, with the covert dose replaced by forward passes of the trained-weight int8 transformer block of Section 5 (one Pythia-6.9B layer on real text activations, quantisation scales applied, the int8 matmuls and fp16 attention alike counted), swept at two shapes that bracket the attention share of the op count. Regressing window energy on covert nominal operations gives a marginal useful cost of 2.54–3.21 pJ/op (both fits $R^2 > 0.999$; one-sided 95% lower limits $[2.53, 3.16]$ pJ/op), against standalone total-quotient one-sided 95% lower limits of $[3.18, 4.22]$ (point means $[3.26, 4.24]$): exactly as in Experiment 4 the amortised idle baseline cancels in the slope, so this is the incremental estimand the useful rungs of Section 5 require, not a total quotient.

Matched-energy witnesses. The corner assumption behind the floor (Assumption 3.1) admits a direct empirical probe, and two measurements bear on it (full protocol in Appendix E). First, matched-pair lower witnesses on one A100 hold the trusted declaration and the observation window fixed by construction: run A executes N declared fp16 iterations on dear operands (the $r^{\mathrm{dec}}_{\mathrm{hi}}$ edge) and idles to a fixed $T = 10$ s window; run B executes the *same* N iterations on cheap operands (the $r^{\mathrm{dec}}_{\mathrm{lo}}$ edge) plus a covert int8 fill tuned to match A's energy, padded to the same window. At $h = 0.10$ the two arms match to $|\Delta E|/E = 0.9\%$ while B hides 3.5×10^4 covert int8 iterations, and the one-sided no-alarm rule of Theorem B.1(ii), set three standard deviations above the honest mean, accepts *every* cheating repeat. The witnesses reach $h = 0.10$, 0.15, and 0.216, where the accepted covert load reaches 0.19, 0.29, and 0.41 machines of declared-format capacity respectively—a measured, capacity- and time-feasible lower bound on the ambiguity. They are lower witnesses, not corner attainment: at $h = 0.216$ the accepted load 0.41 sits well below the model's $\beta(0.216) = 0.72$, and it is the largest load we could get accepted, not a comfortable one (Appendix E). The construction holds only up to a busy-time ceiling $h_{\max} \approx 0.24$, above which run B's work no longer fits the window; above $h_{\max}$ the wall-clock budget binds before the energy budget, so a verifier reading time as well as energy constrains more than the energy channel alone prices, and that joint energy–time inversion is future work. Second, the *transfer gap* behind re-execution (Section 5) is grounded by re-measuring the band edges on three A100s of the same SKU: every edge agrees to 2–3% across devices, so same-SKU re-execution transfers bands at the few-per-cent level.

Assembling the floor. The floor is evaluated inside the jointly feasible fp16-declared scenario of Section 3.2:

the declared band is the fp16 cells' own $[r_{\mathrm{lo}}^{\mathrm{dec}}, r_{\mathrm{hi}}^{\mathrm{dec}}] = [1.10,\ 2.19]$ pJ/op (Experiment 1), the covert cost is Experiment 4's marginal lower limit, the overhead width is $\Delta P_0 = 41.2\,\mathrm{W}$ (Experiment 3), and capacity and γ (Equation (18)) come from the two formats' published peaks. The set is format-consistent by construction—our analysis code rejects any cross-format combination of extrema—and joint attainability of its cheating corner within one observation window is Assumption 3.1. Reading the fp16 cells as the declared band carries one further disclosed assumption: that the declared work is executed in its declared format. No off-site check enforces this; dropping it removes the format pin entirely, so both declared edges open to the full measured envelope and the worst case rises to 1.95 (the bare-meter rung of Section 5). With covert int8 ($\gamma = 2$, $r_{\mathrm{lo}}^{\mathrm{cov}} = 0.51$ pJ/op) the energy-only width at full declared utilisation is

$$\beta_{\mathrm{en}}(1) \;=\; \mathbf{2.39}, \tag{24}$$

or 1.62 with covert work held to fp16 ($r_{\mathrm{lo}}^{\mathrm{cov}} = 0.76$ pJ/op). $\beta_{\mathrm{en}}(1)$ is an energy-only diagnostic, not a physical floor: at full declared utilisation the capacity clip $(1-h)\gamma$ is zero, so no covert work fits and $\beta(1) = 0$. The floor couples the slack to that clip through Equation (20). Figure 7 traces $\beta(h)$ for the two covert scenarios: with covert int8 against declared fp16 the worst case over utilisation is

$$\beta^* \;=\; \mathbf{1.16} \quad \text{at} \quad h^* = 0.42, \tag{25}$$

and with covert work held to fp16 ($\gamma = 1$) it is 0.66 at $h^* = 0.34$. Energy is the binding constraint below the crossing, capacity above it, and both bind at h^*: the worst case sits at intermediate utilisation, where the declared work offers real cover but still leaves much of the machine free.

This peak is a model *upper* bound, and one that no single observation window realises: $h^* = 0.42$ lies above the busy-time ceiling $h_{\max} \approx 0.24$ (Appendix E), so at the headline utilisation the wall-clock budget binds before any schedule reaches the corner. Restricting the maximisation to the utilisations our witness construction can hold ($h \le h_{\max}$) puts the model worst case at $\beta \approx 0.77$; $h_{\max}$ is a property of that construction rather than a general feasibility bound, and a different covert workload would move it (Appendix E). What we *attain* is smaller and measured directly: the witnesses above get $\mathbf{0.41}$ machines of covert int8 work accepted at $h \approx 0.22$. The identified-set width the meter admits is therefore *at least* 0.41 machines—measured, attained, and accepted by the sensor—and *at most* about 1.16 as a model bound whose maximiser is infeasible, or about 0.77 if the bound is confined to a feasible window. Four-tenths of a machine of covert compute passing an honest declaration is a wide identified set for a bare power meter; whether it crosses the whole-machine "vacuity" line is not something the attained witness settles, and we do not claim

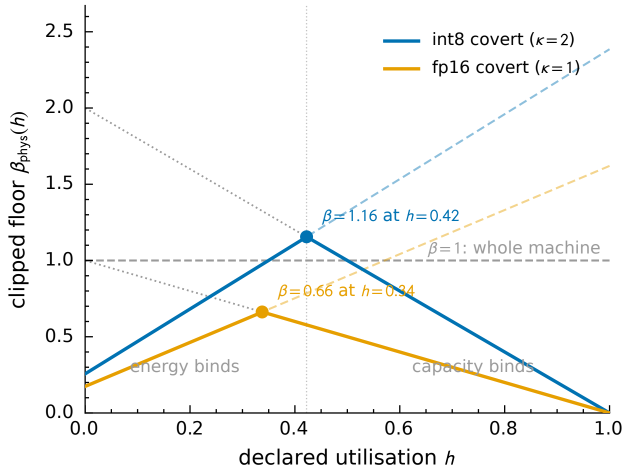


Figure 7. The capacity-clipped floor $\beta(h)$ (Equation (20)) from the jointly feasible fp16-declared bands, for covert int8 ($\gamma = 2$, blue) and covert fp16 ($\gamma = 1$, orange), both priced at Experiment 4's marginal lower limits. Faint dashed lines are the unclipped energy-only widths $\beta_{\mathrm{en}}(h)$ ($\beta_{\mathrm{en}}(1) = 2.39$ and 1.62): energy is the binding constraint left of each crossing, capacity right of it, and both bind at h^*. Dotted lines are the capacity clips $(1-h)\,\gamma$; markers are the worst cases of Equation (21).

it does. The width is set jointly by the energy budget and the silicon, not by either alone.

5. The capability ladder

The bare meter is the worst case: the verifier sees nothing but the power, and the prover has maximum wiggle room. Each capability the verifier gains moves one band edge and shrinks the identified set by a measurable factor. Table 1 and Figure 8 walk down that ladder *cumulatively*: each rung adds one capability to all the rungs above it, and every priced rung is recomputed as the same quantity the headline uses — the capacity-clipped physical worst case $\beta^* = \max_h \beta(h)$ of Equation (21), under the joint model of Section 3.2: fp16-declared case study, covert work priced at the int8 marginal lower confidence limit 0.51 pJ/op (Section 4.3), $\gamma = 2$. The bare rung alone, which pins no declared format, carries the cross-format clip $\gamma - h$ of Section 3.3 in place of $(1-h)\,\gamma$. The energy-only diagnostic $\beta_{\mathrm{en}}(1)$ is reported alongside for orientation only: it is the unclipped no-alarm budget at full utilisation, not a physical floor ($\beta(1) = 0$ — a fully busy machine has no residual capacity to hide work in). Because a verifier that gains a capability can only shrink the set, the priced chain is monotone by construction, and a regression test in the released code enforces it. Two disclosed conditions thread the chain, marked † in Figure 8 and listed per rung in Table 1: *declared-format execution* (from the precision rung down) and an *observed operating point* (from the re-execution rung down). The rest of this section takes the rungs in turn.

verifier gains	type	band edge moved	β^*
nothing (bare meter)	—	measured envelope	1.95
precision declared & checked†	D, C	$r^{\mathrm{dec}} \to [1.10, 2.19]$	1.16
clock & locality observed	T	*unpriced (prospective)*	—
declared work re-executed‡	C	$r^{\mathrm{dec}}_{\mathrm{hi}} - r^{\mathrm{dec}}_{\mathrm{lo}} \to$ gap	0.34
covert work on useful data‡	M	$r^{\mathrm{cov}}_{\mathrm{lo}} : 0.51 \to 2.53$	0.071
covert at top of useful band‡	A	$r^{\mathrm{cov}}_{\mathrm{lo}} \to 3.16$	0.057

Table 1. The capability ladder. Each rung grants the verifier one more capability (cumulative, top to bottom) and is re-priced as the clipped physical worst case $\beta^* = \max_h \beta(h)$ (Equation (21)) under the joint model (fp16-declared case study, int8 covert edge at the Exp. 4 marginal LCB, $\gamma = 2$; the format-unpinned bare rung carries the cross-format clip $\gamma - h$ of Section 3.3); $\beta_{\mathrm{en}}(1)$ is the energy-only diagnostic, not a floor. Numeric band edges in pJ/op. Disclosed conditions: † — assumes the declared work is executed in its declared format (an off-site check cannot enforce this; Section 5); ‡ — additionally assumes re-execution transfers at an observed operating point. Both are inherited by every rung below their first appearance. The type column says what kind of thing each rung is: D — a declaration by the prover, checked off-site; C — offline characterisation on the verifier's own hardware, transferring to the facility only up to a stated gap; T — a trusted on-site observable, here left *unpriced*: attested telemetry does not yet exist, and the measured operand spread w_0 is descriptive evidence, not a certified bound; M — a threat-model restriction (hidden zero-ops are no threat); A — an assumption about the adversary's objective, not a capability.

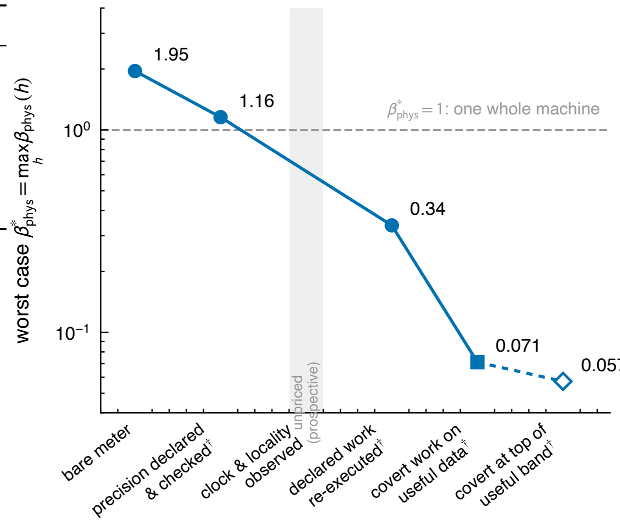


Figure 8. The capability ladder of Table 1 on a log scale; every plotted value is the clipped worst case $\beta^* = \max_h \beta(h)$. Filled circles are verifier capabilities (types D, C); the square rung is forced by the threat model (type M: hidden zero-ops are no threat); the hollow rung is an assumption about the adversary, not a capability (type A). The shaded slot is the unpriced prospective clock/locality rung; daggered labels carry the disclosed conditions of Table 1. The $\beta^* = 1$ line marks one whole machine of conceded ambiguity; the top two rungs are model upper bounds whose maximisers lie above the feasible window, so where they fall relative to that line is not something we claim to settle.

The bare meter ($\beta^* = 1.95$). With power alone the verifier cannot tell declared work from covert, nor one numeric format from another, so both band edges are the full measured envelope of Section 3.3:[6] the honest acceptance ceiling is the dearest measured rate ($r_{\mathrm{hi}} = 19.5$ pJ/op, fp32) and the covert floor is the cheapest incremental operation the silicon admits (the int8 marginal edge, 0.51 pJ/op). No precision is assumed on either side — reading the ceiling as the fp16 cell while opening the floor to int8 would assume the declared format on one edge and deny it on the other, which a bare meter cannot do. The resulting exchange span is $37.9\times$ — wider than the $24.8\times$ measured envelope, because the covert floor here is the int8 *marginal* edge rather than the dearest-to-cheapest ratio of total quotients — wide enough that the capacity clip binds even at low utilisation. The clip is the cross-format $\gamma - h$ of Section 3.3: with no format pinned, the declared operations themselves execute in the cheap covert format and occupy h/γ of resource-time, not h. The clipped worst case peaks at $\beta^* = 1.95$ near $h^* = 0.046$, essentially the int8 density advantage $\gamma = 2$ that a near-idle machine leaves free. This maximiser sits *inside* the feasible window ($h^* < h_{\max} \approx 0.24$), so — unlike the precision rung, whose peak is pushed above the busy-time ceiling — confining the worst case to feasible utilisation does not lower it; the bare-meter figure rests on the corner assumption Assumption 3.1, alongside the measured-scope and covert-edge conditions already flagged. Finally, the rung mixes estimands—a total-quotient ceiling over a marginal covert floor—which makes it a conservative outer sensitivity, wider than an exact single-estimand band would be; the safe direction for a worst case (Appendix D quantifies the mixture).

[6] "Measured" scopes the envelope to the five formats of Section 4.3 (fp32, tf32, fp16, bf16, int8). The A100 also exposes dense int4 (1,248 Top/s) and binary (4,992 Top/s) tensor-core paths (NVIDIA Corporation, 2020) that no committed measurement prices: an int4 covert edge could sit below the int8 marginal edge and carries $\gamma = 4$ against fp16. Unmeasured formats can only widen this rung, so 1.95 is the model worst case within the measured scope and a lower bound on the true full-hardware ambiguity, not a full-hardware worst case.

Declare and check the precision ($1.95 \to 1.16$, conditional). The prover can be made to declare the declared work's numeric format, and the verifier can re-run that work on its trusted cluster and confirm the outputs are what fp16 execution produces. In the taxonomy of Table 1 this rung is a declaration plus an offline check: nothing new is sensed at the facility — and that is exactly its limit. The check makes the declaration *precise*; it cannot make it *enforced*. What the rung buys, collapsing both declared edges from the full measured envelope to the fp16 band's own $[1.10, 2.19]$ pJ/op (the ceiling down from the fp32 rate 19.5, the floor up from the int8 marginal edge 0.51), it buys only under the disclosed condition that the declared work is actually executed in its declared format at the monitored facility († in Table 1); facility-local format enforcement is an open capability, not one this rung grants. Under that condition the rung is precisely the headline case: $\beta^* = 1.16$ at $h^* = 0.42$ (Equation (25)). Here and below, "the verifier knows the precision" always means the *declared* precision. The covert work's precision is unobservable—declaring it would make it not covert—so it never enters the numerator; it resurfaces at the bottom rungs.

Observe the clock and locality (unpriced). Voltage/clock (DVFS, of order 1.5) and locality (weak for a compute-bound kernel) are hardware-wide observables the power meter does not measure; pinning them via other channels would narrow what remains of the declared band down to its operand core. We deliberately quote no number for this rung, for two reasons. First, the capability is prospective on every candidate route: attested clock telemetry (NVML reports the clocks, but firmware-reported numbers count only over the tamper-resistant read path that Section 1 notes does not yet exist), locked clocks by policy, or, in principle, the EM side channel—switching at f radiates at f and its harmonics, and magnetic emanations from a GPU's power delivery are measurable in practice (Maia et al., 2022), though clock-line radiation itself has not been demonstrated—with the voltage then following from the hardware's V–f table. Second, the only number we could put on the rung, the operand core $w_0 = 1.83$, is descriptive within-device evidence: Experiment 2's workload families ran at different achieved clocks (the confound is disclosed in Section 4.3), so w_0 is not a certified population edge and does not price a guarantee. The rung stays in the ladder as the slot a trusted on-site observable would fill — and the next rung depends on it. The locality half is speculative: per-rail metering (core vs. HBM regulators) or interconnect telemetry might resolve it; fortunately Experiment 1 found locality moves r little for compute-bound work.

Re-execute the declared work ($1.16 \to 0.34$, conditional). Re-executing to *measure* the declared work's actual energy per operation, not merely confirm its format, squeezes $r^{\mathrm{dec}}_{\mathrm{hi}} \to r^{\mathrm{dec}}_{\mathrm{lo}}$ and the numerator collapses to a residual set by the *hardware gap*: re-execution runs on the verifier's trusted cluster, not the monitored facility, and energy per operation is hardware-specific—different accelerators, firmware, cooling, operating point—so the measured r^{dec} transfers only up to a gap. That gap, not zero, is the floor on how far re-execution can pin the declared band. The cross-device spread we measured on three A100s of the same SKU—2–3% on every band edge (Section 4.3)—is a lower flavour; across different accelerators, firmware, and cooling the gap is larger and unmeasured. We take the residual gap at a few per cent of the declared rate, ≈ 0.05 pJ/op (an assumption informed by the same-SKU spread, not a cross-hardware measurement), giving $\beta^* = 0.34$ at $h^* = 0.83$ — at this depth the overhead band, not the declared residual, carries most of the width. Read the rung as a sensitivity analysis in the gap, not a measured bound: at the same-SKU spread itself (gap 0.02–0.04 pJ/op) the rung reads 0.29–0.32; at our stated 0.05 pJ/op, 0.34; a cross-vendor gap of 0.2 pJ/op would put it at ≈ 0.54. Measuring that gap — the same declared workload characterised across accelerator generations and vendors — is the experiment this rung waits on.

The rung also inherits a second condition (‡ in Table 1): *re-execution only pays at an observed operating point*. Re-execution measures the declared work's energy per operation at the *verifier's* operating point; if the prover's clock state is unobserved — and the previous rung is exactly the one still unpriced — the prover may run the declared work anywhere on the DVFS curve, and the transfer error is not the hardware gap but the full DVFS factor of ~ 1.5. A verifier with only "power and paperwork" (the power meter plus declarations and re-execution, but no on-site sensing) therefore stalls at $\beta^* \approx 1$: re-executing the declared work buys almost nothing that checking its precision had not already bought.

Restrict covert work to useful data ($0.34 \to 0.071$). This rung is forced by the threat model rather than granted by a channel, and the logic runs in steps. Covert work is unseeable *by definition*: no declaration reaches it, so its floor $r^{\mathrm{cov}}_{\mathrm{lo}}$ is bounded only by the silicon — the int8 marginal edge, 0.51 pJ/op. But the cheapest operations on that edge are operand zeros, and zeros compute nothing useful: hidden zero-ops are no threat. We therefore measured the *incremental* cost of useful covert work directly: an int8-quantised transformer block built from one trained Pythia-6.9B layer, run on real text activations (not synthetic patterns), with the quantisation scales applied and every metered operation — the int8 matmuls and the fp16 attention alike — counted. Sweeping that block as the covert dose inside the Experiment-4 construction (a fixed declared burst and fixed window, so baseline power cancels in the dose slope; Appendix C) prices genuinely useful covert work at a *marginal* $r_{\mathrm{useful}} = [2.53, 3.16]$ pJ/op (one-sided 95% lower limits; the standalone total quotient E/C is $[3.18, 4.22]$ (one-sided lower limits), dearer only because it amortises idle power the covert work does not draw). Restricting the adversary to work worth doing raises the covert edge to the bottom of that band, $0.51 \to 2.53$ pJ/op, while the declared-side residual stays at the hardware gap — the covert band's own width prices nothing on the declared side — and the rung drops to $\beta^* = 0.071$: every slack joule now buys the adversary $\approx 4.9\times$ fewer useful operations than silicon-floor accounting would concede. Because this edge is an incremental dose slope, not a standalone total quotient, it is a genuine marginal lower bound on the covert cost — the estimand the denominator requires (Appendix C), measured the same way as Experiment 4's int8 covert edge.

Covert work at the top of the useful band ($0.071 \to 0.057$). The final rung is an assumption, not a capability. Only by assuming the covert work forgoes all of its remaining freedom inside the useful band and pays that band's ceiling, $r_{\mathrm{lo}}^{\mathrm{cov}} \to 3.16$ pJ/op, does the residual fall to ≈ 0.057. A bound of that order — covert compute at several per cent of capacity — exists only on this rung, conditional on an assumption about the adversary's objective rather than on any measurement; the distinction matters for the power-monitoring layers proposed in the verification literature (Baker et al., 2025).

6. Discussion

The meter is an anchor, not an instrument. The result reads pessimistic — against a bare power meter a matched-energy witness demonstrably hides four-tenths of a machine of covert compute, and the model worst case runs higher still, to $\beta^* = 1.95$ against a bare meter — a figure that rests on the joint feasibility of the band corners (Assumption 3.1) — but the ladder inverts the reading. The meter's role in a verification regime is not to count operations by itself; it is to be the one channel whose integrity needs no trust in the prover, the anchor against which declarations, attested clocks, and re-execution are checked. Power plus checked declarations reaches $\beta^* \approx 1.2$; adding an observed operating point and re-execution of the declared work reaches ≈ 0.34, and restricting covert work to useful data ≈ 0.071 of capacity. Every rung is a protocol requirement a treaty could stipulate, together with a disclosed condition it rests on, and Equation (21) states exactly which band each requirement moves and by how much. The negative result is equally operational: because re-execution transfers only at the verifier's operating point, "power plus paperwork" — declarations and re-execution with no on-site sensing — stalls at $\beta^* \approx 1$, a regime design worth ruling out before it is proposed.

What no verifier capability reaches. The denominator of Equation (17) is a statement about the adversary, not the instrumentation. Covert work is by definition undeclared, so its exchange rate is bounded only by the silicon and by what the threat model says the adversary wants: excluding zero operands (which compute nothing) raises the covert floor from the int8 marginal edge to the cheapest *useful* rate — a measured $4.9\times$ step, 0.51 to 2.53 pJ/op — and the remaining $1.25\times$, the useful band's own width, closes only by assuming the adversary declines cheap-but-useful work — sparse activations, quantised weights — that our own measurements show exists. Any bound quoted for power-based verification should state which side of that assumption it stands on.

From one device to a facility. Our bands are device physics, measured on one A100; a facility is a sum of devices behind shared power delivery and cooling, and we deliberately do not offer a fleet-average version of Equation (17). Scaling the floor up is a *multi-resource allocation problem*, not a change of units: each device carries its own capacity clip and utilisation, the prover chooses how declared work is spread across devices (and formats) to maximise cover, and the facility meter adds cooling and conversion overheads that the prover partially controls. The identified set is then the projection of a per-device product of constraints, and its width depends on the allocation the prover is free to pick; treating h as a single fleet average can under- or overstate it. What does carry over qualitatively: overhead bands widen at facility scale, and cheaper datapaths widen the precision envelope beyond our $24.8\times$ and raise the capacity ratio γ of the cheapest format—int4 and binary tensor-core paths on the A100 itself (unmeasured here; Section 5), FP8 and FP4 on newer parts. Making the multi-device floor precise—including which per-device telemetry restores the single-device analysis—is open.

Limitations. Seven caveats bound the claims. (i) The bands come from one vendor, one GPU model, one driver, and one cluster; the ladder's numbers are A100-specific, though the closed form is not. (ii) The band edges are read from a factorial sweep whose corner cell (int8, zero operands) is directly measured, but on synthetic operands; genuinely useful covert work prices at the measured marginal useful band $[2.53, 3.16]$ pJ/op instead (Experiment 5's dose-response; the standalone total-quotient lower limits are $[3.18, 4.22]$), and the bottom rungs depend on that band's width. (iii) The hardware gap behind the re-execution rung (≈ 0.05 pJ/op) is an assumption. The same-SKU piece is grounded—band edges agree to 2–3% across three A100s (Section 4.3)—but the cross-hardware gap (different accelerators, firmware, cooling) remains unmeasured. (iv) DVFS was never swept — the platform denied clock-locking — so the voltage factor rides on literature numbers rather than our own; persistence mode was likewise fixed on, so its contribution to the controllable idle range went unmeasured. (v) Energy was read from on-die telemetry in a non-adversarial lab characterisation (Section 4.3); the counter carries no published GA100 accuracy specification, so the $\pm 5\%$ tolerance we apply is adopted from NVML's documented Fermi/Kepler figure, and the ~ 100 ms telemetry cadence is an empirical characterisation (Yang et al., 2023), not a vendor specification; an off-chip meter would add its own calibration band, which only widens β. (vi) On shared nodes, windows stalled by co-tenants were dropped by a pre-specified power guard, and high-tail windows are flagged rather than deleted; all raw rows remain on disk. The flagged rows are excluded from the published bands and regressions—including the fp16 declared band and the marginal-dose fits, not only the descriptive medians—and with five repeats per cell the 20% trim always removes one repeat, excursion or not. The sensitivity is small: restoring every positive flagged record moves the fp16 band to $[1.10, 2.20]$ pJ/op, the int8 marginal lower limit to 0.510, and the headline to $\beta^* = 1.162$. The quantities that actually enter the bound are of three kinds, and we state the status of each rather than claim one statistical coverage guarantee over all of them. (a) The covert denominator—the one quantity a lower bound is genuinely sensitive to—is a one-sided 95% lower confidence limit on a regression slope (Experiments 3 and 4, and the useful-band dose slope), coverage-controlled by construction. (b) The declared band $[r^{\mathrm{dec}}_{\mathrm{lo}}, r^{\mathrm{dec}}_{\mathrm{hi}}]$ enters only as a *difference* of per-cell-median edges at a common saturated rate (Appendix C), so the amortised baseline cancels and it acts as a band *width*, not a point estimate needing its own coverage; its target population is the achievable per-shape operating points of the declared format on the audited A100 SKU, and the edges are that observed operating envelope; used as design-time security edges they remain sampled operating points, not certified physical extrema. (c) The overhead width ΔP_0 and the capacity/γ inputs are conservative worst-case *design* parameters, not sampled estimates: ΔP_0 is the observed idle envelope stretched by the adopted $\pm 5\%$ meter tolerance (widening it only widens β), and $C_{\max}$ and γ are published hardware peaks. We therefore do not claim simultaneous statistical coverage over the rectangle these edges span; the rectangle is a deliberately conservative over-approximation, as no single schedule attains its combined extrema (Assumption 3.1). (vii) The locality contrast is attenuated in the fp32/tf32 cache-resident cells, whose operands marginally spill L2.

What richer telemetry adds. Everything here is time-integrated, and the sufficiency of total energy for a question about totals is a consequence of our *free-variation* model: because $r(t)$ and $P_0(t)$ may vary arbitrarily within their bands, the trace's shape carries no information about totals beyond its integral. Real hardware is not perfectly free—power-ramp limits, DVFS governor dynamics, the instantaneous clip $F(t) \leq F_{\max}$, and any *declared schedule* the prover commits to all constrain how the trace can move, and a verifier who models them can extract more from the resolved trace than from its integral. Conditional on the free-variation model, then, the time *profile* adds nothing about how much compute ran; unconditionally, that is an open modelling question, not a theorem. The profile also carries different information — workload cadence, training-step periodicity — that bears on what *kind* of compute ran rather than how much; that question needs different machinery and we do not treat it here.

The same caveat applies beyond the time axis. Our result prices the integrated-energy channel under free variation; a verifier with richer instrumentation is outside the model. Multiple simultaneous channels (per-rail power, electromagnetic, thermal), cryptographically authenticated telemetry (the proof-of-learning and zero-knowledge proof-of-training protocols of Assumption 2.2), or independently controlled measurements the prover cannot shape (locked clocks, tamper-resistant attested counters) each carry constraints the integral does not, and any of them may narrow the identified set below the widths reported here. The impossibility we state is for total-energy sensing under the free-variation threat model, not for analogue sensing in general.

7. Conclusion

A bare power meter cannot count operations. The energy cost of an operation is unknown and data-dependent—across the five formats we measured the envelope spans $24.8\times$ on an A100 (0.79–19.5 pJ/op), with the hardware's unmeasured int4 and binary paths wider still, and even inside one jointly feasible declared format the band and the marginal covert cost leave an energy slack ($\beta_{\mathrm{en}}(1) = 2.39$, Equation (24), an energy-only diagnostic since no covert work fits at full utilisation). Coupling that slack to silicon capacity gives the clipped floor of Equation (20). What we *attain* and the sensor accepts is a matched-energy witness hiding 0.41 machines of covert int8 work at $h \approx 0.22$; the model worst case peaks higher, $\beta^* = 1.16$ at $h^* = 0.42$ (Equation (25), 1.95 against the bare meter with the declared format unenforced, within the measured five-format scope), but that peak is an upper bound whose maximiser sits above the busy-time ceiling—confined to a feasible window the model worst case for the precision-checked sensor is about 0.77. The identified-set width the meter admits is thus at least 0.41 machines and at most about one, set jointly by the energy budget and the silicon. This is not a statistical shortfall a better meter or a longer window repairs; it is a structural degeneracy of the integrated-energy forward map under free variation — a property of the model, not of the A100. The specific widths above (the $24.8\times$ span, $\beta_{\mathrm{en}}(1) = 2.39$, the measured 0.41, and the model $\beta^* = 1.16/1.95$) are contingent A100 estimates, scoped to the formats we measured; the closed form they instantiate is device-agnostic. An identified set this wide—four-tenths of a machine of covert compute demonstrably passing an honest declaration—substantially weakens the raw joules-to-FLOPs conversion in the governance literature against an adversary free to vary energy intensity.

But the floor is a closed form built from named physical bands, and that is what makes it useful: each verifier capability moves one band edge by a measured amount, and every rung is re-priced as the same clipped worst case $\max_h \beta(h)$. Precision declared and checked brings the measured-scope worst case from 1.95 to 1.16, conditional on the declared work executing in its declared format; re-execution of the declared work at an observed operating point to ≈ 0.34 — a residual that rests on a stated hardware-gap assumption, not a measurement (Table 1). The rungs interlock — re-execution pays only at an observed operating point, so power plus paperwork stalls at ≈ 1 — and no capability reaches the covert side: restricting covert work to the measured useful-operand band brings ≈ 0.071, and the final collapse to ≈ 0.057 closes only by assumption on the adversary. This paper prices each rung, with its conditions stated, so that a verification regime can be designed — or rejected — on numbers rather than on the intuition that joules imply FLOPs.

References

Ansari, S. Hardware-level governance of AI compute: A feasibility taxonomy for regulatory compliance and treaty verification, April 2026. URL https://arxiv.org/abs/2604.04712.

Arefin, S. E. and Serwadda, A. Exploiting HDMI and USB ports for GPU side-channel insights, October 2024. URL https://arxiv.org/abs/2410.02539. Published at ACM CODASPY 2025.

Baker, M., Kulp, G., Marks, O., Brundage, M., and Heim, L. Verifying international agreements on AI: Six layers of verification for rules on large-scale AI development and deployment, July 2025. URL https://arxiv.org/abs/2507.15916.

Chandrakasan, A. P., Sheng, S., and Brodersen, R. W. Low-power CMOS digital design. *IEEE Journal of Solid-State Circuits*, 27(4):473–484, 1992. doi: 10.1109/4.126534.

Chatterjee, N., O'Connor, M., Lee, D., Johnson, D. R., Keckler, S. W., Rhu, M., and Dally, W. J. Architecting an energy-efficient DRAM system for GPUs. In *2017 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, pp. 73–84, 2017. doi: 10.1109/HPCA.2017.58.

Chaudhuri, A., Shukla, S., Bhattacharya, S., and Mukhopadhyay, D. "Energon": Unveiling transformers from GPU power and thermal side-channels, August 2025. URL https://arxiv.org/abs/2508.01768.

Choukse, E., Warrier, B., Heath, S., Belmont, L., Zhao, A., Khan, H. A., Harry, B., Kappel, M., Hewett, R. J., Datta, K., et al. Power stabilization for AI training datacenters, August 2025. URL https://arxiv.org/abs/2508.14318. Microsoft.

Clark, S. S., Ransford, B., Rahmati, A., Guineau, S., Sorber, J., Xu, W., and Fu, K. WattsUpDoc: Power side channels to nonintrusively discover untargeted malware on embedded medical devices. In *2013 USENIX Workshop on Health Information Technologies (HealthTech 13)*. USENIX Association, 2013.

Gangwal, A., Piazzetta, S. G., Lain, G., and Conti, M. Detecting covert cryptomining using HPC, 2019. URL https://arxiv.org/abs/1909.00268. Published at CANS 2020.

Gao, Y., Qiu, H., Zhang, Z., Wang, B., Ma, H., Abuadbba, A., Xue, M., Fu, A., and Nepal, S. DeepTheft: Stealing DNN model architectures through power side channel, September 2023. URL https://arxiv.org/abs/2309.11894. Published at 2024 IEEE Symposium on Security and Privacy (S&P).

Garg, S., Goel, A., Jha, S., Mahloujifar, S., Mahmoody, M., Policharla, G.-V., and Wang, M. Experimenting with zero-knowledge proofs of training. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2023. doi: 10.1145/3576915.3623202.

Heim, L. and Koessler, L. Training compute thresholds: Features and functions in AI regulation, 2024. URL https://arxiv.org/abs/2405.10799.

Horowitz, M. 1.1 computing's energy problem (and what we can do about it). In *2014 IEEE International Solid-State Circuits Conference (ISSCC) Digest of Technical Papers*, pp. 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323.

Hu, X., Liang, L., Li, S., Deng, L., Zuo, P., Ji, Y., Xie, X., Ding, Y., Liu, C., Sherwood, T., and Xie, Y. DeepSniffer: A DNN model extraction framework based on learning architectural hints. In *Proceedings of the 25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pp. 385–399. ACM, 2020.

Jia, H., Yaghini, M., Choquette-Choo, C. A., Dullerud, N., Thudi, A., Chandrasekaran, V., and Papernot, N. Proof-of-learning: Definitions and practice. In *2021 IEEE Symposium on Security and Privacy (SP)*, 2021. doi: 10.1109/SP40001.2021.00106.

Latif, I., Newkirk, A. C., Carbone, M. R., Munir, A., Lin, Y., Koomey, J., Yu, X., and Dong, Z. Single-node power demand during AI training: Measurements on an 8-GPU

NVIDIA H100 system. Technical report, Office of Scientific and Technical Information (OSTI), 2025. URL https://www.osti.gov/biblio/2555926.

Maia, H. T., Xiao, C., Li, D., Grinspun, E., and Zheng, C. Can one hear the shape of a neural network? Snooping the GPU via magnetic side channel. In *31st USENIX Security Symposium (USENIX Security 22)*, pp. 4383–4400. USENIX Association, 2022.

Monfared, S. K., Ganji, F., Holcomb, D., and Tajik, S. Timing and memory telemetry on GPUs for AI governance, 2026. URL https://arxiv.org/abs/2602.09369.

Naghibijouybari, H., Neupane, A., Qian, Z., and Abu-Ghazaleh, N. Rendered insecure: GPU side channel attacks are practical. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pp. 2139–2153. ACM, 2018.

NVIDIA Corporation. NVIDIA A100 tensor core GPU architecture. Whitepaper, 2020. URL https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/nvidia-ampere-architecture-whitepaper.pdf. Dense tensor-core peaks: 19.5 TFLOP/s fp32, 156 TFLOP/s tf32, 312 TFLOP/s fp16/bf16, 624 TOPS int8.

O'Gara, A., Kulp, G., Hodgkins, W., Petrie, J., Immler, V., Aysu, A., Basu, K., Bhasin, S., Picek, S., and Srivastava, A. Hardware-enabled mechanisms for verifying responsible AI development. In *ICML 2025 Workshop on Technical AI Governance (TAIG)*, 2025. URL https://arxiv.org/abs/2505.03742.

Pallipadi, V. and Starikovskiy, A. The ondemand governor: Past, present, and future. In *Proceedings of the Linux Symposium*, volume 2, pp. 223–238, 2006.

Patel, P., Choukse, E., Zhang, C., Goiri, Í., Warrier, B., Mahalingam, N., and Bianchini, R. POLCA: Power oversubscription in LLM cloud providers, 2023. URL https://arxiv.org/abs/2308.12908.

Rahman, R. and Tajdari, S. Detecting hidden ML training with zero-overhead telemetry, 2026. URL https://arxiv.org/abs/2606.19262.

Reuel, A., Bucknall, B., Casper, S., Fist, T., Soder, L., Aarne, O., Hammond, L., Ibrahim, L., Chan, A., Wills, P., Anderljung, M., Garfinkel, B., Heim, L., Trask, A., Mukobi, G., Schaeffer, R., Baker, M., Hooker, S., Solaiman, I., Luccioni, A. S., Rajkumar, N., Moës, N., Ladish, J., Bau, D., Bricman, P., Guha, N., Newman, J., Bengio, Y., South, T., Pentland, A., Koyejo, S., Kochenderfer, M. J., and Trager, R. Open problems in technical AI governance, 2025. URL https://arxiv.org/abs/2407.14981.

Sastry, G., Heim, L., Belfield, H., Anderljung, M., Brundage, M., Hazell, J., O'Keefe, C., Hadfield, G. K., Ngo, R., Pilz, K., Gor, G., Bluemke, E., Shoker, S., Egan, J., Trager, R. F., Avin, S., Weller, A., Bengio, Y., and Coyle, D. Computing power and the governance of artificial intelligence, February 2024. URL https://arxiv.org/abs/2402.08797.

Seferis, E. and Fist, T. Detecting compute structuring in AI governance is likely feasible. Working paper, 2025. URL https://openreview.net/forum?id=qseqw1sWzz.

Shavit, Y. What does it take to catch a Chinchilla? Verifying rules on large-scale neural network training via compute monitoring, 2023. URL https://arxiv.org/abs/2303.11341.

Tang, B. J., Chen, Q., Weiss, M. L., Frey, N., McDonald, J., et al. The MIT Supercloud workload classification challenge, 2022. URL https://arxiv.org/abs/2204.05839.

The White House. Executive order 14110: Safe, secure, and trustworthy development and use of artificial intelligence. Federal Register 88 FR 75191, 2023. URL https://www.federalregister.gov/documents/2023/11/01/2023-24283/.

The White House. Executive order 14148: Initial rescissions of harmful executive orders and actions. Federal Register Doc. 2025-01901, 2025. URL https://www.federalregister.gov/documents/2025/01/28/2025-01901/.

Wei, J., Zhang, Y., Zhou, Z., Li, Z., and Al Faruque, M. A. Leaky DNN: Stealing deep-learning model secret with GPU context-switching side-channel. In *2020 50th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, pp. 125–137. IEEE, 2020.

Xu, H., Gong, C., Liu, B., Zheng, H., Chen, B., and Li, M. Wave: Leveraging architecture observation for privacy-preserving model oversight. In *Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2026. doi: 10.1145/3779212.3790247.

Yang, Z., Adámek, K., and Armour, W. Part-time power measurements: nvidia-smi's lack of attention, 2023. URL https://arxiv.org/abs/2312.02741.

A. From the CMOS power law to the exchange rate

Section 2.1 sketches the route from the CMOS power law to the exchange rate; the back-of-the-envelope argument of Section 4.2 additionally adopts the factored model Equation (23). We give the derivation in full here and locate the terms the factorisation drops. Resolving the static term of Equation (1), total chip power separates into a dynamic (switching) part and a static part,

$$P = P_{\text{dyn}} + P_{\text{stat}}, \qquad P_{\text{dyn}} = \alpha\, C\, V^2 f, \qquad P_{\text{stat}} = V I_{\text{leak}} + P_{\text{fixed}}, \tag{26}$$

with α the activity factor, C the switched capacitance, V the supply voltage, f the clock, $V I_{\text{leak}}$ the leakage power, and P_{fixed} the cooling, I/O, and idle draw. The forward model Equation (2), $P = rF + P_0$, is slope–intercept in the operation rate F, so the exchange rate is the *marginal* energy per operation and the overhead is the intercept,

$$r = \frac{\partial P}{\partial F}, \qquad P_0 = P\big|_{F=0}. \tag{27}$$

Experiment 3 estimates Equation (27) directly, regressing mean power on sustained nominal-op rate with idle cells as the $F = 0$ anchors (Section 4.3).

The dynamic term gives r. The operation rate is $F = N_{\text{op}}\, f$, where N_{op} is the number of nominal operations the datapath retires per cycle at the operating format (an idle unit lowers the activity α, not the clock f). At a fixed format the clock is thus the knob through which the operation rate moves, $\partial f/\partial F = 1/N_{\text{op}}$, and the marginal rate of Equation (27) follows by the chain rule, holding α, C, and V fixed,

$$r \;=\; \frac{\partial P_{\text{dyn}}}{\partial F} \;=\; \frac{\partial P_{\text{dyn}}}{\partial f}\,\frac{\partial f}{\partial F} \;=\; \frac{\alpha\, C\, V^2}{N_{\text{op}}} \;=\; \alpha \left(\tfrac{C}{N_{\text{op}}}\right) V^2, \tag{28}$$

and the clock drops out: energy per operation does not depend on how fast the operations run, only on what each switching event costs and how many operations it delivers. The rate is constant in F, so $P_{\text{dyn}} = rF$ is linear through the origin, exactly the affine form the forward model Equation (2) assumes and Experiment 3 fits. The capacitance switched per operation, C/N_{op}, has two structural sources: the arithmetic datapath, whose width is set by the numeric format, and the data-movement path, set by where the operands live and which kernel runs. Factoring it accordingly,

$$\frac{C}{N_{\text{op}}} \;=\; \kappa(\text{precision})\; C_{\text{eff}}(\text{locality, kernel}), \tag{29}$$

with κ collecting the format scale (datapath width and the operation packing N_{op}) and C_{eff} the locality and kernel structure, and substituting into Equation (28), gives the model Equation (23). This is a re-parameterisation by physical origin, not a unique factorisation: κ, α, and C_{eff} are all switching quantities, grouped by cause—format, operand data, memory locality—so that each maps to a single verifier category.

Short-circuit sits in r, inside the switched capacitance. Chandrakasan et al. (1992) also carry a *short-circuit* term for the direct-path current that flows while an input transitions and the NMOS and PMOS networks briefly conduct together. It is triggered by the same switching events as P_{dyn}, so it scales with the operation rate and survives the marginal derivative of Equation (27). In a well-designed circuit it is a small fraction of the switching power, so we fold it into the effective switched capacitance C_{eff} of Equation (29): it lands in r, but introduces no new verifier category.

Leakage sits in P_0, not r. The static power P_{stat} of Equation (26) flows whether or not the machine computes, so $\partial P_{\text{stat}}/\partial F = 0$: by Equation (27) it contributes nothing to the marginal rate r and is entirely the intercept P_0, where the derivation of Equation (10) already places it and carries it as a bounded nuisance band. A leakage-per-operation term $r_{\text{leak}} = P_{\text{stat}}/F$ would appear only if r were defined as the *average* energy per operation, P/F, which mixes in the intercept and is operating-point dependent; we keep the marginal definition Equation (27), consistent with Equation (2), and so carry no separate leakage term in r.

B. What β is a worst case over

The derivation of Section 3 moves through four quantities that are easy to conflate, so we separate them formally here. Throughout, the verifier observes the total energy E, knows the declaration C_{dec} (hence h), and carries the band knowledge

$\bar{r}^{\rm dec} \in [r^{\rm dec}_{\rm lo}, r^{\rm dec}_{\rm hi}]$, $\bar{r}^{\rm cov} \geq r^{\rm cov}_{\rm lo}$, $E_0 \in [E_{0,\rm lo}, E_{0,\rm hi}]$ of Section 3.1, together with the capacity constraint Equation (19). The bands are treated as a rectangle—each edge attainable independently of the others—and that the cheating corner of the rectangle is itself attainable is Assumption 3.1.

Proposition B.1 (Four estimands)**.** *Fix $C_{\rm dec}$ and the no-alarm rule Equation* (14)*. Then:*

(i) Identified width, given the observation. *For an observed total E, the C-projection Equation* (6) *of the identified set has width $W(E) = \min\big[\max\big(0,\ (E - r^{\rm dec}_{\rm lo} C_{\rm dec} - E_{0,\rm lo})/r^{\rm cov}_{\rm lo}\big),\ (1-h)\,\gamma\,C_{\max}\big]$: the covert load the trace cannot rule out depends on how much energy was actually drawn. The outer clamp to 0 covers observations below the honest lower edge $r^{\rm dec}_{\rm lo} C_{\rm dec} + E_{0,\rm lo}$ (see (ii)): there no honest-plus-covert schedule reproduces E, the identified set is empty, and $W(E)$ is 0 by convention rather than negative.*

(ii) One-sided acceptance region (intentionally permissive). *The no-alarm rule Equation* (14) *accepts exactly the totals $E \leq r^{\rm dec}_{\rm hi} C_{\rm dec} + E_{0,\rm hi}$. The rule is deliberately permissive: covert work only* adds *energy, so the test guards only against draws that run* high *and does not certify that every accepted draw is honestly explainable. Accepted draws split in two. Those at or above the honest lower edge $r^{\rm dec}_{\rm lo} C_{\rm dec} + E_{0,\rm lo}$* are *explainable by honest declared work at some admissible rate and overhead; those* below *it are accepted too but are* not *honestly explainable—honest work cannot draw that little—so their honest identified set is empty ($W(E) = 0$ in (i)). Such under-draws indicate* under-*delivery, which lies outside the covert-compute threat model, so the rule tolerates them by design rather than imposing a lower alarm. The worst-case widths (iii)–(iv) are set at the region's* upper *boundary, where the identified set is largest, so the empty lower regime never binds them. The modelled decision rule, and the empirical 3σ rule of Section 4.3, are therefore one-sided.*

(iii) Worst case over accepted observations. *Write $W_{\rm en}(E)$ for the energy-only width in (i) before its capacity clamp (the first argument of the* $\min$*). The largest such width over the acceptance region, in units of $C_{\max}$, is attained at the region's top edge:* $\sup_E W_{\rm en}(E)/C_{\max} = \beta_{\rm en}(h)$ *of Equation* (17)*.*

(iv) Capacity-clipped minimax width. *Restoring Equation* (19)*, the worst case becomes $\beta(h) = \min[\,\beta_{\rm en}(h),\ (1-h)\,\gamma\,]$ of Equation* (20)*, and the verifier-free headline is its maximum over utilisation, β^* at h^* (Equation* (21)*).*

Under Assumption 3.1 the suprema in (iii) and (iv) are attained; otherwise they are upper bounds.

The proof is the derivation of Section 3: (i)–(ii) restate Equations (14) and (15); (iii) is the corner argument of Section 3.1 evaluated at the top of the acceptance region; (iv) intersects with Equation (19). The distinctions matter because only (i) is a statement about a particular audited window; (iii) and (iv) are design-time guarantees about what *any* accepted window could hide. Quoting (iii) as if it were (i) overstates the ambiguity of a typical window; quoting it without (iv) overstates what physics admits.

C. What the bound requires of the measurements

Appendix A derives the affine forward map Equation (2) from device physics. The security bound does not need it. This appendix states the weaker condition it does need, and why the two band edges are estimated by different means.

Two roles, two estimands. The declared edges enter Equation (16) only through their *difference* $r^{\rm dec}_{\rm hi} - r^{\rm dec}_{\rm lo}$ at a common saturated rate, so any baseline amortised equally into both edges cancels and a measured total quotient E/C is the correct estimand for the numerator (the overhead enters once, separately, as ΔE_0). The covert cost enters as the *denominator* $r^{\rm cov}_{\rm lo}$, where nothing cancels: the bound is a valid security guarantee only if $r^{\rm cov}_{\rm lo}$ is a genuine one-sided *lower* bound on the *incremental* covert energy. Write the true draw as $P = g_x(F_{\rm dec}, F_{\rm cov})$, some device-dependent function of what the silicon executes, with the subscript x standing for the execution details the verifier never sees. The condition is then

$$g_x(F_{\rm dec}, F_{\rm cov}) - g_x(F_{\rm dec}, 0)\ \geq\ r^{\rm cov}_{\rm lo}\, F_{\rm cov}, \tag{30}$$

for every admissible execution g_x. Under only this incremental condition the honest energy slack of Equation (16) still divides by $r^{\rm cov}_{\rm lo}$, so the floor holds without the global affine map of Equation (2); the affine model is the special case $g_x = P_0 + r^{\rm dec} F_{\rm dec} + r^{\rm cov} F_{\rm cov}$, which additionally makes $\beta_{\rm en}(h)$ linear in h. We therefore identify $r^{\rm cov}_{\rm lo}$ by a dose–response intervention (Experiment 4 for the generic int8 edge, Experiment 5 for the useful-operand edge), not by fitting a power–rate curve; a total-quotient covert edge is inadmissible here, and every covert edge the bound uses is a marginal dose slope.

D. Two conservative accounting choices, quantified

Two accounting choices in the model bound are deliberately adversary-favorable; this appendix quantifies what each concedes.

Published peaks vs. achieved rates. The capacity clip Equation (19) is evaluated at the *published peak* capacities rather than the rates our kernels achieve. Peaks give the covert stream more room than it can really use, so the conceded β is larger than an achieved-rate evaluation would give. On our own measured peaks (246.9 Top/s fp16 and 307.7 Top/s int8, so $\gamma = 1.25$ rather than 2) the bare-meter rung falls from 1.95 to 1.22 and the precision rung from 1.16 to 0.91. Assumption 3.1 asks for achieved throughputs when it certifies that a particular *witness* is realisable in one window, and the witnesses of Section 4.3 are checked that way (Appendix E); the model bound itself is deliberately evaluated at peaks.

The estimand mixture on the bare-meter rung. The bare-meter rung of Section 5 prices its covert floor marginally while its acceptance ceiling is a total quotient. The marginal int8 edge (0.515 pJ/op) is a third cheaper than the cheapest total quotient the hardware shows (0.786 pJ/op), and the overhead term divides by it, so $\Delta P_0/(r_{\text{lo}}^{\text{cov}} F_{\max}) \approx 0.26$ is half again the 0.168 a total-quotient covert edge would give: under the estimand the bound actually requires (Appendix C), the overhead band carries more of the width. The mixture also runs through the numerator—the ceiling is a total quotient whose amortised baseline does not cancel against the marginal floor, and the overhead band is then carried once more as its own term—so the rung is a conservative outer sensitivity, wider than an exact single-estimand band would be, which is the safe direction for a worst case.

E. The matched-energy witness protocol

The matched-pair witnesses of Section 4.3 hold the trusted declaration and the observation window fixed by construction: run A executes N declared fp16 iterations on dear operands and idles to a fixed $T = 10$ s window; run B executes the same N iterations on cheap operands plus a covert int8 fill tuned to match A's energy, padded to the same window. This appendix records the protocol details and checks behind the accepted loads quoted in the body (0.19, 0.29, and 0.41 machines at $h = 0.10$, 0.15, and 0.216).

The energy match. Over eight interleaved repeats at $h = 0.10$, run A drew 1291.3 ± 4.4 J and run B 1279.3 ± 5.6 J, matching to $|\Delta E|/E = 0.9\%$ while B hid 3.5×10^4 covert int8 iterations behind the one-sided 3σ no-alarm rule of Theorem B.1(ii).

Clock matching. Clock locking is denied on the platform (Section 4.3), so the arms are matched by cadence rather than by a held clock: both reach the same 1410 MHz compute clock, and the per-window *median* clock differs between them only through the idle-pad share, exactly as in Experiment 4.

The power-cap asymmetry is conservative. Run A's dear burst draws ≈500 W against run B's ≈245 W and consequently sits at the card's software power cap, which truncates A's measured honest energy; since the acceptance threshold is set from A's mean, and the covert dose is tuned against it, the cap tightens the bar run B must clear and understates the covert load the tuner can afford.

Achieved rates: lower witnesses, not corner attainment. The achieved incremental rates are 1.84 pJ/op for the dear declared arm (below the band ceiling 2.19), 0.82 for the cheap declared arm, and 0.54 for the covert fill (above the marginal edge 0.51): the witnesses meet the shared constraints of Assumption 3.1 but do not jointly reproduce the rectangle's extreme edges, which is why the accepted load 0.41 at $h = 0.216$ sits below the model's $\beta(0.216) = 0.72$.

The margin at the topmost witness. The acceptance margin narrows as the witness load rises, and the topmost witness is the thin one: at $h = 0.216$ the cheating arm's spread is $2.6\times$ the honest arm's, and its dearest repeat (1969.5 J) lands above the honest arm's dearest (1964.6 J), clearing the 3σ threshold by 0.86% of the honest mean. The 0.41 figure should be read as the largest load we could get accepted, not a comfortable one.

The busy-time ceiling. Run B multiplexes cheap declared work and roughly two covert operations per declared operation into one window, so above $h_{\max} \approx 0.24$ its work no longer fits T. The ceiling is a property of this construction, not a general feasibility bound: it is set by this kernel's achieved rates and covert dose ratio, and includes a 10% protocol margin so the idle pad never underflows (without it, $h_{\max} = 0.27$ and the restricted worst case of Section 4.3 is 0.82 rather than 0.77). A different covert workload would move it.

F. Benchmark protocol, hardware, and provenance

Experiment 1 (Section 4.3) reads the *total energy quotient* E/C off a single A100 by holding the workload fixed and moving one factor of Equation (23) at a time. The quotient is the band-edge estimand of Section 4.3—not the marginal rate $r = \partial P/\partial F$ of Equation (2), which Experiments 3 and 4 estimate as regression slopes—and we keep the two apart throughout. Both declared band edges are read at the same saturated rate, so their amortised baselines cancel in the band-edge difference the floor uses; the $\approx$5% spread in achieved rate across cells leaves a small residual, which widens the slack conservatively. This appendix gives the hardware, provenance, and mechanics: the workload and why its operation count is exact, how each driver is set, and how the energy is metered. Experiments 2 and 3 reuse the same protocol with a different swept axis; Experiment 4 reuses its metering inside fixed-length composite windows (declared burst + covert dose + idle pad).

Hardware and provenance. Experiments 1–4 ran sequentially on one NVIDIA A100-SXM4-80GB (driver 595.84), in a single Slurm allocation on the OzSTAR facility on 22 July 2026 (340, 135, 85, and 70 recorded windows respectively). The released records and capture scripts index experiments by their capture identifiers, which differ from the numbering used here—in particular, capture `exp4` is the cross-device sweep and *not* Experiment 4, whose records are `exp7`; the repository's `README` maps the two schemes. Every record carries the device UUID, sweep id, configuration snapshot, and git commit, and the band-extraction code refuses to mix devices silently; because Experiment 4's marginal cost and Experiment 1's declared band price the same floor, the guard requires them to share one device. Seven cards of one SKU appear in total, by UUID prefix: `GPU-ac1cbae9` (Experiments 1–4); `GPU-4bb93dfb`, `GPU-617f2717`, `GPU-aa412071` (cross-device sweep); `GPU-757ff02a` (the trained-weight total quotient, reported only as a contrast to the marginal edge) and `GPU-747045fa` (Experiment 5, the incremental useful-work edge that prices the bottom two rungs of Section 5); `GPU-108c1549` (the matched-energy witnesses of Section 4.3). Each single-device band is therefore compared against declared edges measured on a different card, and the same-SKU spread (2–3% on every band edge; Section 4.3) bounds the device variation it inherits. Full manifests accompany the released records; the trained weight/activation tensors themselves are not committed, but the manifests record their SHA-256 hashes and the GPU pre-flight validation metrics, so the extraction is checkable against the release but not replayable from the repository alone.

Operating point. The platform denied clock locking and power capping, so DVFS is a recorded axis rather than a controlled one: each window logs the governor-selected SM clock (achieved 1335–1410 MHz on the band sweeps), the bands are stratified into 45 MHz clock strata, and no window was flagged with a *harmful* throttle (thermal or hardware power brake). The densest operand bursts do sit at the card's software power cap by design, which truncates their measured power and is therefore conservative for a dearest-rate edge.

Meter accuracy and cadence. `nvmlDeviceGetTotalEnergyConsumption` exposes a cumulative millijoule register integrated in firmware. NVML publishes no accuracy specification for this counter on GA100; the documented $\pm 5\%$ applies to the power readings of the earlier Fermi and Kepler generations, and we adopt that figure as a working meter tolerance wherever one is needed, always in the direction that widens the bound. The counter's underlying power telemetry updates on a $\sim$100 ms cadence and samples only part of the runtime (Yang et al., 2023); each measured cell therefore fills a $\sim$1.5 s window ($\gg$100 ms), as described under *Metering the energy* below.

The workload and its exact operation count. Every cell runs a dense square matrix multiply $A\,B$ with $A \in \mathbb{R}^{M\times K}$, $B \in \mathbb{R}^{K\times N}$ (we take $M = N = K = n$). Each of the MN output entries is a dot product of length K—K multiplies and $K-1$ adds—so by the standard two-operations-per-multiply-add convention the matmul costs

$$C \;=\; 2MNK \;=\; 2n^3 \quad \text{op}, \tag{31}$$

counted as nominal operations of whatever format the cell runs (int8 multiply–adds included; Section 2.1), known *analytically* from the shapes alone, with no profiler or hardware counter involved. A measured window runs `iters` such matmuls back to back, so its nominal compute is $C_{\text{win}} = 2n^3$ `iters` and the energy per operation is $r = E_{\text{win}}/C_{\text{win}}$. Because C is exact under the nominal convention (the literal scalar count is $MN(2K-1)$, within 0.03% of $2MNK$ at the band cells' $K=2048$), r inherits the meter's accuracy directly; and because r is intensive (energy *per* operation), cells of different size and `iters` compare directly.

Setting the swept drivers.

- **Precision** κ is the tensor `dtype`: `fp32`, `tf32` (fp32 operands routed through the tf32 tensor-core path), `fp16`, `bf16`, and `int8` (an integer-matmul kernel, int8×int8→int32). Lower precision charges less capacitance per multiply-accumulate, lowering κ.
- **Locality** C_{eff} is operand *residency* at a byte-identical shape, not problem size: sweeping it by changing n would change the operation count and confound the comparison, so instead we fix the shape at $n=2048$ and vary how many distinct operand sets the window rotates through iteration to iteration. The A100 has a 40 MB L2; at $n=2048$ an fp16 A–B operand set is $\sim$16 MB, so:
  - *resident* (one operand set, reused every iteration): the working set fits the L2, so most traffic is served on-chip and HBM activation is avoided: low C_{eff};
  - *streamed* (eight operand sets, round-robined): a reuse is evicted before it recurs, so every iteration streams fresh tiles from HBM, engaging its arrays, peripheral logic, and I/O: high C_{eff}.

  Both arms run the identical shape, kernel, and operation count Equation (31); only the memory-traffic path differs, so locality no longer confounds the count. The kernel stays compute-bound in both arms—a 2048^3 matmul is high arithmetic intensity—so streaming raises C_{eff} without making the workload memory-bound, and Figure 3 accordingly shows locality moves r only slightly.

The third driver, supply voltage V^2 (DVFS), we could not sweep: the platform denied user clock-locking, so the clock was left to the governor and merely recorded per run. The operand *values* are a fourth axis: Experiment 1 crosses all six operand distributions (Appendix G) with precision and locality in a full factorial, so the pooled envelope already contains the operand spread; Experiment 2 re-runs the operand axis alone, at fixed precision and locality, as the descriptive operand-only map of Appendix G.

Metering the energy. Energy is read from the on-die NVML total-energy counter: the window energy is one subtraction $E_1 - E_0$, with a `cuda.synchronize()` fence bracketing each latch so the delta spans exactly the window's kernels; where the counter is unavailable we integrate the power query in a background sampler. Each cell fills a $\sim$1.5 s window with matmuls run back to back on pre-allocated buffers, with no host traffic inside the window: long enough that counter quantisation is negligible, and saturated enough that the joules are the datapath doing arithmetic, not allocation or transfer. A few warm-up windows absorb JIT and clock ramp, several measured windows follow, and the per-cell median is kept; shuffling the cell order and periodically re-measuring a fixed reference cell guards against slow thermal drift. The largest per-cell median over the sweep is the band ceiling r_{hi}; Figure 3 plots the full set.

G. How Experiment 2 varies the operands

Experiment 2 (Section 4.3) holds the kernel, shape, and precision fixed and moves *only the operand values*. Every configuration runs the identical matmul $A\,B$ with the identical operation count Equation (31), so the energy differences track the switching-activity factor α of Equation (23)—how many bits physically toggle in the datapath as the multiply–accumulates stream through—read descriptively, per the clock confound disclosed in Section 4.3. A datapath that sees the same bits over and over charges little capacitance; one whose bits flip on every operation charges the most. The six distributions are constructed to walk α from its floor to its ceiling. Below we show each on a 4×4 patch; the real operands are the same patterns at full size, generated by `operand_fill` in `powertoflops/bench/workload.py`.

1. Zeros—the activity floor. Every entry is zero, so no bits ever change and α is at its minimum:

$$A_{\text{zeros}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is the cheapest run at fixed fp16 and sets the descriptive fp16 operand floor, 1.16 pJ/op as a total quotient; combined with the int8 format in Experiment 1's factorial it reaches the measured envelope floor $r_{\mathrm{lo}} = 0.786$ pJ/op. (The floor's covert denominator is Experiment 4's marginal cost, not these quotients; Section 4.3.)

2. Low-entropy—a few distinct levels. Entries are drawn from only four values $\{0, 1, 2, 3\}$. With so few distinct bit patterns, few bits differ between neighbouring operands, so operand-bit entropy—and hence switching—stays low:

$$A_{\text{low-entropy}} = \begin{pmatrix} 1 & 0 & 3 & 1 \\ 2 & 2 & 0 & 3 \\ 0 & 1 & 1 & 2 \\ 3 & 0 & 2 & 1 \end{pmatrix}.$$

3. Structured-sparse—half the operands forced to zero. Dense random values, but a fixed half of the columns are deterministically zeroed (every other column), giving exactly 50% zeros in a regular pattern: the structured sparsity real accelerators exploit, rather than random dropout. The persistent zero columns roughly halve the switching relative to a fully dense operand:

$$A_{\text{sparse}} = \begin{pmatrix} 0 & 0.4 & 0 & -0.9 \\ 0 & -0.7 & 0 & 0.2 \\ 0 & 0.8 & 0 & 0.5 \\ 0 & -0.3 & 0 & -0.6 \end{pmatrix}.$$

4. Dense-random—generic busy data. Every entry is an independent uniform draw from $[-1, 1]$: no zeros, full-width mantissas, high but not maximal switching. This is the realistic-workload reference:

$$A_{\text{dense}} = \begin{pmatrix} 0.31 & -0.82 & 0.55 & -0.14 \\ -0.67 & 0.09 & -0.48 & 0.73 \\ 0.21 & 0.64 & -0.95 & 0.38 \\ -0.52 & -0.27 & 0.86 & -0.41 \end{pmatrix}.$$

5. Declared-typical—the realistic-activation stand-in. Every entry is an independent standard-normal draw, $\mathcal{N}(0, 1)$: a proxy for the activations and weights a genuine declared workload streams, with full-width mantissas and no engineered structure. Its switching sits alongside dense-random—high but not maximal—and it is the operand class that characterises the honest-window energy scatter used by the declared-work model:

$$A_{\text{decl}} = \begin{pmatrix} 0.42 & -1.13 & 0.27 & -0.68 \\ -0.35 & 0.91 & -1.42 & 0.11 \\ 1.06 & -0.24 & 0.58 & -0.83 \\ -0.72 & 0.39 & -0.16 & 1.24 \end{pmatrix}.$$

6. Adversarial-toggle—the activity ceiling. Engineered for maximal switching: the matrix is a checkerboard of two *complementary* bit patterns, `0x5555...` and its bitwise inverse `0xAAAA...`, so adjacent operands differ in *every* bit (maximal Hamming distance) and the operand path sees maximal input-level switching per operation. For floats the pattern is built in an integer bit-view of the same width and reinterpreted as the float type, so the raw bits—not the numeric value—are controlled exactly. Writing the two patterns as p and $\bar{p}$:

$$A_{\text{adv}} = \begin{pmatrix} p & \bar{p} & p & \bar{p} \\ \bar{p} & p & \bar{p} & p \\ p & \bar{p} & p & \bar{p} \\ \bar{p} & p & \bar{p} & p \end{pmatrix}, \qquad p = \texttt{0x5555}\ldots, \;\; \bar{p} = \texttt{0xAAAA}\ldots$$

A matmul forms each output as a dot product $\mathrm{out}[i, j] = \sum_k A[i, k]\, B[k, j]$, streaming $k = 0, 1, 2, \ldots$ and loading one operand into the multiplier's input register each cycle. Because every row and every column of the checkerboard alternates $p, \bar{p}, p, \ldots$, that register walks

$$\underbrace{\texttt{0101 0101}}_{k=0} \to \underbrace{\texttt{1010 1010}}_{k=1} \to \underbrace{\texttt{0101 0101}}_{k=2} \to \cdots$$

flipping *all* its bits on *every* cycle. Dynamic CMOS energy is paid per bit-transition, so this maximises the switching the operand registers can drive: a w-bit operand toggles w bits per cycle, against $\approx w/2$ for dense-random (two random words differ in about half their bits) and 0 for zeros or any constant operand (the register never changes). The checkerboard is simply the 2-D layout that makes the row sequence (feeding A) and the column sequence (feeding B) both alternate this way at once, for every output entry.

The energy-per-operation spread from the cheapest patch (A_{zeros}) to the densest operands (dense-random, declared-typical, and the adversarial checkerboard), at fixed operation count, is the observed operand spread $w_0 = 1.83$ of Figure 4. The measured ordering follows the textbook one, zeros $<$ low-entropy $<$ structured-sparse $<$ dense-random $\approx$ declared-typical $\approx$ adversarial-toggle. Per the clock confound of Section 4.3, the spread is descriptive within-device evidence, not a certified population edge—and maximal input-level toggling does not itself certify maximal switching inside the Tensor Core internals, whose operand encodings are opaque (nor do zero operands mean no hardware bits switch); no security bound in the paper rests on it.